



The immune system of marine invertebrates: Earliest adaptation of animals[☆]

R. Bharathi Rathinam^a, Arpit Acharya^a, A. Jancy Robina^a, Husne Banu^{b,*}, Gayatri Tripathi^{a,**}

^a ICAR- Central Institute of Fisheries Education, Mumbai, India

^b ICAR- Central Institute of Freshwater Aquaculture, Bhubaneswar, India

ARTICLE INFO

Keywords:

Ocean invertebrates
Symbiotic microbes
Pattern recognizing receptors
Cellular and humoral immunity

ABSTRACT

The present article provides insights into the immunity of ocean invertebrates. Research in this field has revealed hallmark components and strategies of the animal immune system, which have both theoretical and practical utility. From phylum Porifera to arthropods, animal immunity has evolved over various timescales. Within these timescales, dynamic environmental factors and microbial influences have shaped invertebrate immunity in an adaptive manner, resulting in the acquisition of complex genetic traits. These genetic traits play a crucial role in coding for multiple components, including antigens binding to cellular and cytoplasmic receptors, antimicrobial peptides, proteases, and proteins in pioneer animals. Marine invertebrates, such as sponges, cnidarians, mollusks, echinoderms, and crustaceans, exhibit diverse innate immune mechanisms influenced by their environment and microbial interactions. Despite lacking adaptive immune responses, marine invertebrates rely primarily on innate immunity. Their immune system encompasses physical and chemical barriers that prevent pathogen entry, as well as specialized cells and humoral factors that combat infections and minimize potential damage. In summary, this review explores the immunity of ocean invertebrates, highlighting key discoveries related to both cellular and humoral defenses across various marine species.

Introduction

The origin of life is believed to be from water. Around 3.5 billion years ago, the first animal was born inside the ocean. Since then, animals have gone through a long-term evolution of life. The early group of animals had different body features and physiology compared to today's animal on earth. The evolution has changed their way of living in accordance with dynamic life on earth. Their body features and physiology has been changed to adapt to the extremities of nature and acclimatize with a particular environment. Some of them are still surviving in the ocean. Apart from their use in nutritious food supplements, traditional medicine, recreation and aesthetic enjoyment, sea animals have a great role in biodiversity, climate control and discoveries of many intellectual facts such as insights on regeneration, embryogenesis, reproductive biology, wound healing properties and in advanced animal physiology etc.

Over 155 years ago only the marine animals such as sea urchins and starfish were chosen to be the favorable animal to analyze

embryogenesis and fertilization owing to their large number of externally fertilized eggs [1]. The role of neuronal control in locomotion was long back studied in rock Lobster [2] which is useful for Parkinson related disease treatment. Sharks and skates are being used for kidney and reproductive biology related studies [3,4]. Owing to their resistance to genetic instability and wound healing property, long lived white sharks are found useful to study cancer and age-related diseases [5]. Apparently, all these animals have primitive body systems but are not inferior as compared to advanced animal physiology. The immune systems of such animals are fascinating and house of many informative discoveries. A strong immune system is essential to win the struggle for life and apparently ancient groups of animals also carry such defense mechanisms. Sea animals have been selected for milestone discoveries in animal immunology. Famous Russian scientist Mechnikov in 1882 has studied tiny starfish larvae and discovered some moving cells, later named phagocytes which can scavenge the fungal pathogens. Looking at these vast ranges of ocean animals and their interesting life style, the paper represents a review on the defense system of such animals.

[☆] Husne Banu & Gayatri Tripathi should be considered as joint corresponding authors.

^{*} Corresponding author: ICAR- Central Institute of Freshwater Aquaculture, Bhubaneswar, Odisha, India.

^{**} Corresponding author: ICAR- Central Institute of Fisheries Education, Mumbai, Maharashtra-400061, India.

E-mail addresses: husnebanu786@gmail.com (H. Banu), gayatri1267@gmail.com (G. Tripathi).

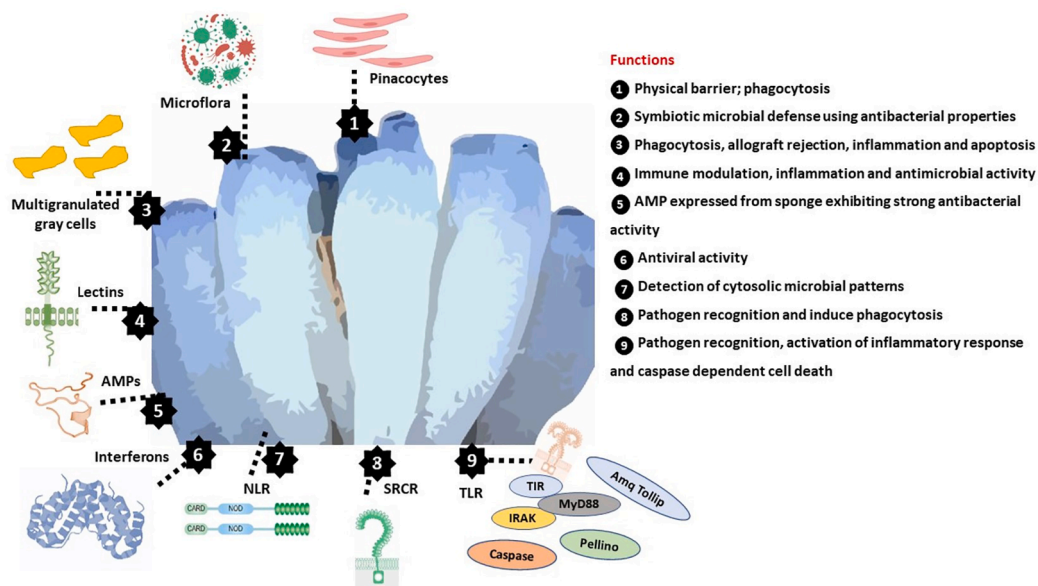


Fig. 1. Immune molecules involved in poriferan immunity and their possible functions.

Immunity of non-bilaterian

Porifera

The name implies the body surface of sponges covered with porocytes. These are most primitive multicellular benthic animals which lack many of the complex traits i.e., neurons and gonads, nevertheless it can accomplish different functions similar to other animals. They filter feeds on a heterotrophic group of bacteria but simultaneously it can distinguish symbiotic groups of microbes from others [6]. Approximately 40 % of sponge biomass could carry specific group of unique microbial consortia known as sponge symbionts which includes Proteobacteria, Actinobacteria, Nitrospira, Acidobacteria, Chloroflexi and Poribacteria [7,8]. Sponges are classified into low microbial abundance and high microbial abundance based on the magnitude of 2–4 orders of microbial load. The bioactive metabolites derived from symbionts provide chemical defensiveness, competitiveness, acclimatization and adaptation features against biotic and abiotic stress in sponges. The defensiveness is not only in the sense of protecting against predation and spatial competition among benthic communities but also immunity against non-symbiotic microbes.

The polybrominated diphenyl ethers isolated from *Dysidea granulosa* show a range of broad-spectrum antibacterial activity against both gram positive and negative bacteria [9]. The wound activated conversion of phopho-calyculin to highly toxic calyculin in *Discodermia calyx* is an example of chemical defense imparted by the symbiotic filamentous bacteria *Candidatus Entotheonella* towards its host in distress [10]. Similarly, some of the polyketides namely macrolide swinholide and theopalauamide are potent antifungal compounds derived from *Theonella swinhoei* [11]. Apart from these, carotenoid 3R, 3'R-zeaxanthin, derived from *Reniera japonica* associated *Cytophaga* species have a potential role against reactive oxygen species [12]. Not only chemical defense, the microbiome also carries functional traits related to prokaryotes defense mechanism i.e., CRISPR (clustered regularly interspaced short palindromic repeats), restriction modification, DNA phosphorothioation and phage growth limitation systems [13]. The metagenomic analysis of three Mediterranean sponge microbiomes has revealed that such an abundance of immune related functional traits not only protects their existence but provide passive immunity to the sponges from bacteria, virus and phages [14].

The symbiotic microbial community is vulnerable to various biotic

and abiotic factors i.e., temperature, acidification, pollution etc. The microbial shift or damage indirectly makes sponges prone to diseases and dominated with various opportunistic bacteria [15]. Hence apart from symbiotic microbial support sponge do have its own immune mechanisms similar to higher animals. The sponge dermal layer is the first physical barrier against microbial entry which is composed of a single delicate layer of pinacocytes. Phagocytosis has been noticed in the exopinacocytes of *Ephydatia fluviatilis* [16] when exposed to latex beads. Existence of allorecognition has long been studied in marine sponges which mean there is a well-defined mechanism for self and non-self-differentiation. Formation of multi-granulated gray cells in response to the allografting is one of the findings implying presence of immune cells in sponges [17].

The extracellular and cytosolic cell receptor associated with various antigen binding has also been detected from sponges. The scavenger receptor cysteine-rich (SRCR) domains are one of the important groups of aggregation receptors characterized in sponge which are known to be involved in pathogen binding and phagocytosis [18]. There are 39 bioactive lectins isolated from sponges which have various roles in immune modulation, inflammatory and antimicrobial activity [19]. The lectins isolated from marine sponge *Chondrilla caribensis* are known to curb biofilm formation by implementing its carbohydrate recognition domain [20].

The toll like receptors (TLR) are one of the most conserved and important mechanism of metazoan immune system including the sponges. A large number of immune transcripts had been identified and their expression was detected in TLR pathway immune activation of sponges i.e., Toll/interleukin-1 (IL-1) receptor (TIR) domain adaptor IgTIRs, myeloid differentiation factor 88 (MyD88), Toll interacting protein (AmqTollip) and Pellino [21]. Their presence in such early stage indicates importance of differentially activated TLR signaling pathway which includes adaptor binding and expression of antimicrobial compounds for overall development and antimicrobial defense. Additionally, the transcripts homologous to three families i.e., Toll/interleukin-1 receptor/resistance (TLR family), er/Thr/Tyr kinase domain (IRAK family), and CASC (caspase family) has also been reported from demo-sponge *Suberites domuncula*. The *in-situ* hybridisation shows outer layer of sponge i.e., Exo-pinacoderm and endo-pinacoderm and cells lining the aquiferous canal specifically express maximum number of such immune transcripts and that implies their strong role in distinct antigen trafficking [22].

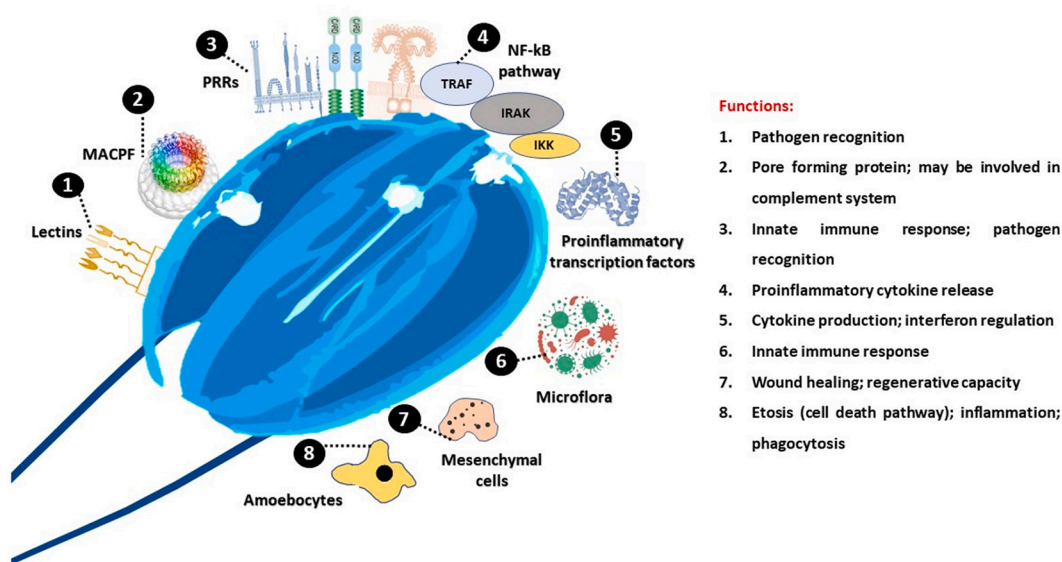


Fig. 2. Immune molecules involved in poriferan immunity and their possible functions.

Apart from TLRs, a large number of transcripts ($n = 135$) of intracellular Nucleotide binding and Oligomerization Domain like receptors has also been detected in *Amphimedon queenslandica* [23]. By the implementation of a diverse group of adaptors, effector and signaling molecules, the sponge engages the NLRs for detection of various cytosolic microbial patterns [24]. Such a diversity in TLRs and NLRs (NOD like receptors) perhaps acquired from vertical transmission of innate immune memory. However, the symbiotic cross talk between bacteria and sponge has an invincible role in molding the innate immune players. Comparative metatranscriptomics of two sponges namely *Stylissa carteri* and *Xestospongia testudinaria*, classified as low microbial abundance (LMA) and high microbial abundance groups (HMA) has revealed a maximum number of TIR domains reported from HMA group whereas the absence of such repertoire was compensated by expansion of maximum number of NLR and SRCR (Scavenger receptor cysteine rich) like domains in LMA sponge [25]. This indicates the microbial symbiosis not only provides direct defense functionality but it also molds the innate defense of animals by passing on the acquired memory of pathogen recognition receptors.

The immune molecules associated with poriferan immunity and their possible functions were depicted in Fig. 1. Overall, the microbes associated with sponges produce bioactive metabolites providing chemical defenses, such as antibacterial and antifungal compounds. Furthermore, sponges also have cellular and humoral immune defenses, including phagocytosis and allrecognition, and possess antigen-binding receptors like scavenger receptor cysteine-rich (SRCR) domains and lectins. Toll-like receptors (TLRs) and NOD-like receptors (NLRs) are crucial for pathogen detection and immune signaling in sponge immunity. The microbial community is vulnerable to environmental stressors, and its damage can make sponges prone to diseases. Microbial symbiosis influences the innate immune system of sponges, potentially passing on acquired immune memory, with immune receptor diversity and functionality varying between low and high microbial abundance sponges.

Ctenophora

Phylum Ctenophora (also known as 'Comb jellies') encompasses a group of exclusively marine and bilaterally symmetrical invertebrates with an intricate ciliary locomotion system. The phylum constitutes two different classes namely Tentaculata which includes Hormiphora (The sea walnut), Pleurobrachia (The sea gooseberry), Ctenoplana, Cestum (The Venus' Girdle) and Nuda which includes Beroe, Mnemiopsis etc.

Ctenophores, like all basal metazoans, rely exclusively on the innate immune system to combat adverse conditions and pathogens. A wide array of immune related genes is believed to be involved in a delicate pattern to constitute the tritecta of innate immune response: allrecognition, downstream signaling and effector responses. However, the number of dedicated studies to comprehend the immune system profile of Ctenophores is minimal. Since the phylum occupies a basal position in the metazoan phylogenetic tree, the immunity related studies will be useful in mapping out the evolutionary aspects of innate immunity over the years.

The existence of a diverse repertoire of putative immune related genes has been revealed in the meager immune related research studies in the phylum Ctenophora. The presence of 56 scavenger receptors, 2 peptidoglycan recognition proteins, interferon regulatory factor (IRF), and proteins containing a membrane attack complex component/perforin (MACPF) domain in *Pleurobrachia bachei* [26], lectins and complement factor-b in *Mnemiopsis leidyi* [27] provide strong evidential support to the existence of a sophisticated and ancient innate immune system similar to other basal metazoans. Two putative TLR homologs, a putative P2/ MPEG-1 homolog containing both the MACPF domain and P2 domain, a single NLR homolog with sequence similarity to human NLR3 and several putative homologs of NF-κB pathway such as TRAF (Tumor necrosis factor receptor (TNF-R)-associated factor), IRAK (Interleukin-1 receptor-associated kinases), and IKK (Inhibitor of nuclear factor-κB (IκB) kinase) have been identified in *M. leidyi* and *Hormiphora californiensis*. The existence of proinflammatory transcription factors has been revealed with the identification of a single putative IRF homolog, a single putative Fos protein and Jun protein which are involved in the heterodimerization of functional activator protein-1 (AP-1) in *M. leidyi* and *Hormiphora californiensis*.

Furthermore, a mobile population of amoebocyte-like cells capable of phagocytosis has been identified in the mesoglea of *M. leidyi* [28]. The possibility of innate immune priming has also been explored in the ctenophore *M. leidyi* where the differential immune gene expression pattern following homologous and heterologous bacterial re-exposure was studied [27]. A study by [29] revealed the existence of a bacterial consortium associated with the ctenophores *M. leidyi* and *Beroe ovata*. As with other basal metazoan sister taxon, the bacterial symbiont consortium is expected to play an influential role in the innate immune response of Ctenophores, however, the underlying mechanisms are still unknown. The immune cells of *M. leidyi* were found to possess the function of Etosis which is a cell death pathway characterized by the

formation of extracellular traps (ETs) [30]. This finding shows that the mechanism of ETosis is ancient and conserved along the evolutionary lineage, starting from basal invertebrates to advanced higher vertebrates.

As a primary response to injury, the mesenchymal cells of the mesoglea move to the surface of injury and cover the edges of the wound with spike-shaped filopodia originating from the epithelial cells [31]. This marks the sophistication of regenerative capability in these basal metazoans. The existence of a complementary relationship between the primitive immune system and the regeneration ability confers an evolutionary advantage to these basal metazoans. More studies are required to comprehend the intricate roles of the immune proteins in the immune cascade and subsequent effector responses in the phylum Ctenophora.

The immune molecules so far reported to be associated with Ctenophora immunity and their possible functions were depicted in Fig. 2. In summary, these Ctenophora rely solely on the innate immune system, featuring genes involved in allorecognition, signaling and effector responses. Key immune components identified include scavenger receptors, peptidoglycan recognition proteins, interferon regulatory factors and MACPF domain proteins. Notable species, such as *Pleurobrachia bachei* and *Mnemiopsis leidyi*, exhibit sophisticated immune systems with TLR homologs, NF- κ B pathway components and proinflammatory transcription factors. The bacterial symbiont consortium associated with ctenophores likely influences their immune response. Additionally, ctenophores exhibit significant regenerative capabilities, with mesenchymal cells moving to injury sites, enhancing their evolutionary fitness.

Cnidaria

Phylum cnidaria establishes itself as the pioneer of functional tissue-level organizational development among metazoans. The cnidarians are the early ancestors of meta organism community concept harboring an intricate complex of microbiome (both symbiome and pathobiome). However, the adaptive immune system has failed to establish itself as the dominant player in Cnidarian immunity, mirroring the primitive immune mechanism of all other invertebrates. In addition to the conventional allorecognition and downstream effector cascade of innate immunity, Cnidarian innate immune system is equipped with the ability to accommodate symbiotic relationships with algal and bacterial symbionts as seen in corals and sea anemones. Phylum cnidaria encompasses five classes: Anthozoa, Cubozoa, Hydrozoa, Scyphozoa, and Staurozoa. The class Anthozoa comprised of hard corals, soft corals, sea pens, gorgonians and sea anemones). Other classes such as Cubozoa consist of box jellies; Hydrozoa includes hydras, hydroids, hydromedusae and siphonophores; Scyphozoa comprised of true jellyfish), and Staurozoa includes stalked jellyfish.

Coral immunity

Similar to the poriferan ancestors, a major patron of coral immunity is the microbial community associated with them. There exists a dynamic relationship between corals and the symbiotic holobiont which is mostly species specific and varies according to the environmental stimuli i.e., temperature, salinity, pollution and radiation [32]. Apart from nutrient cycling, photosynthesis and wound activated chemical defense, associated microbes have a direct impact on host susceptibility to pathogenic bacteria via antibiotic production. The 20 % of coral isolated bacteria from mucus of *Acropora palmata* express antibiotics which regulate growth of pathogenic bacteria [33]. Rajasabapathy et al. [34] has also reported 52 isolates of bacteria from marine sponge, displaying antimicrobial activity against single or multiple bacterial strains. The *Pseudoalteromonas* sp. isolated from many coral strains display strong antibacterial activity against pathogenic bacteria [35,36]. But often the homeostasis of corals and symbiont is being jeopardized by

various factors like sudden disease outbreaks, pollution, and coral bleaching events that directly or indirectly influence the immune system of the host organism. A healthy *Montastraea annularis* harbors more of *Pseudomonas* sp. which is replaced by pathogenic *Vibrio* sp. in case of bleached corals [37]. Analysis of microbes associated with *Acropora millepora* has also revealed that *Pseudoalteromonas* and *Vibrio coralliilyticus* related ribotypes were dominant in sloughed coral mucus whereas Gammaproteobacteria were dominant in healthy corals [38].

To combat the invading pathogenic microbes, corals do have their own immunity. Owing to its multifaceted roles against different stress agents, the coral immunity can be categorized as an immune response against abiotic factors (temperature, light, pH) and biotic factors (pathogens). Thermal stress response in corals recruits an array of genes encoding heat shock proteins (Hsps) i.e., Hsp 70 and 90 [39], cell-cell adhesion molecules, apoptosis regulators and proteins involved in calcium ion homeostasis [40,41]. The antioxidants Cu-Zn superoxide dismutase (Cu-Zn SOD) act as scavengers of oxygen radicals that are released by the thermally stressed symbionts. The physical barriers constitute the first line of defense against biotic stress agents. In corals, the physical barriers are formed by the mucus and sclerite layers that are present within the ectoderm. However, the physical barriers are breached when the biotic stress overwhelms the defense system. The ectodermal layer carries symbiotic microbes and antimicrobial peptides (AMPs). The symbionts efficiently prevent the pathogens from colonization by competitive exclusion. Antimicrobial peptides constitute the chemical barriers and the first characterized scleractinian AMP, Damicornin, was found to be expressed by ectodermal cells of *Pocillopora damicornis* in response to a non-pathogenic stimulus. The AMP exhibited antimicrobial activity against Gram-positive bacteria and the filamentous fungus, *Fusarium oxysporum* but was ineffective against Gram-negative bacteria including *Vibrio coralliilyticus* [42]. Another AMP (AmAMP1), characterized in *Acropora millepora*, was found to be active against both Gram-positive and Gram-negative bacteria [43]. More investigational studies are required in aspects of identification and characterization of AMPs in corals.

The coral mucus associated beneficial bacteria secretes antibiotics which can restrict the entry of opportunistic microbes. The mucus in turn modifies pathogenicity of these beneficial bacteria. The mucus which is isolated from *A. palmata* in the summer season does not possess antimicrobial properties which suggests that cell signaling molecules of mucus alters irrespective of species and geographical region, according to the environment stimuli. Several immunological secondary metabolites have also been identified as immune response components. The polycyclic diterpenes from the gorgonian coral *Briareum polyanthas* [44], caucanolides A – F (1 – 6) [45] and oxapolycyclic diterpenoid, named bipinnapterolide B [46] isolated from extracts of *Pseudopterogorgia bipinnatathe*, crude extracts from the gorgonian coral *Leptogorgia virgulata*, likely containing homarine [47], and a multitude of uncharacterized antimicrobial agents has been also documented in over a dozen members of the Plexauridae, Gorgonidae and Ellisellidae families [48,49].

The wound repair process of corals is a cascade of four step events having a great role in preventing secondary tissue infection. The first step involves coagulation due to the release of transglutaminase carrying granular cells and the second step is mediated by the acidophilic granular amoebocytes which circulate through the mesoglea connective tissue. Amoebocytes have been implicated in melanization, phagocytosis, and inflammatory response of corals [50,51]; Mydlarz et al., 2008). Further fibroblasts infiltrate and proliferate at the lesion site of infection to form an epidermal-like layer. Following the re-epithelialization process, the final maturation of the epithelial layer occurs along with scar tissue development which completes the wound healing process in corals. The proPO mediated antigen inactivation is a highly conserved mechanism which is also detected in corals. Upon PRR mediated detection of antigens the activated PO is produced which induce melanin and intermediate quinone production and both are toxic

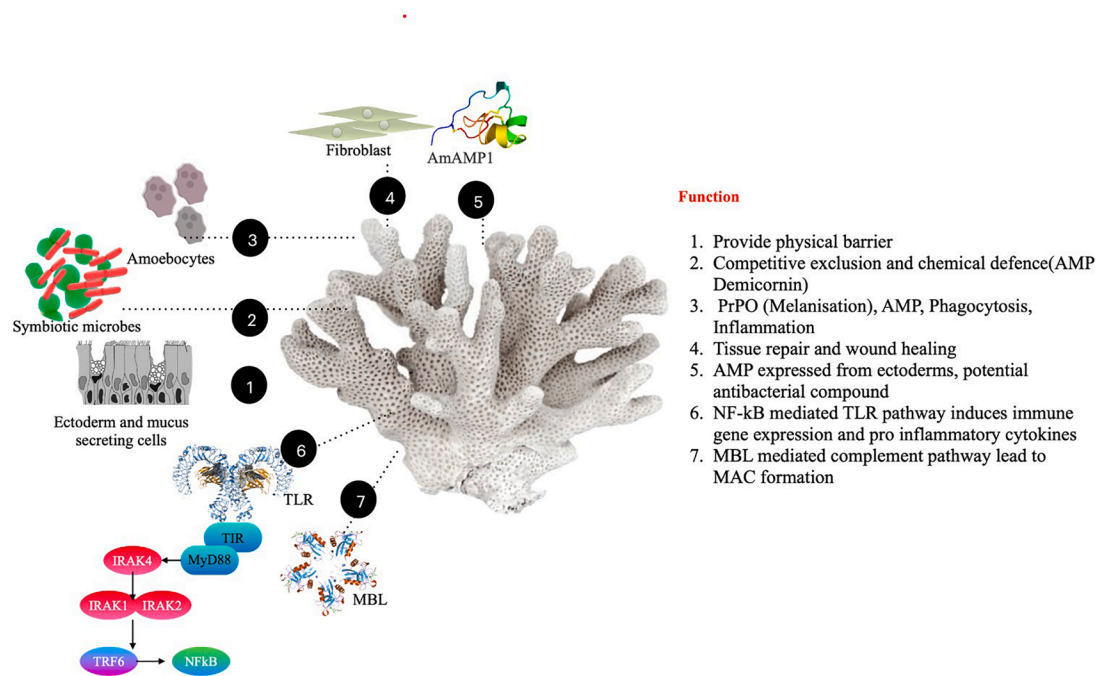


Fig. 3. Molecules of coral immune system and their possible function.

to antigens. The granules of amoebocytes contain PO and ROS which qualify these cells as important agents of cellular immune response [52]. The accumulation of granular amoebocytes at the injury site is accompanied by an increase in melanization response, upregulation of phagocytic activity [52], and secretion of immune reactive enzymes and anti-fungal metabolites. This constitutes an integral part of the cellular immune cascade of corals. The study also highlighted the varying levels of amoebocyte aggregation between the soft alcyonacean corals and the hard scleractinian corals. Amoebocyte accumulation was found to be comparatively lower in hard corals [53,54] than the soft corals warranting prospective research studies in the variable expression of amoebocytes across the coral community.

The TLRs and NLRs signalling via the JNK (Jun N-terminal kinase), MAPK (Mitogen-activated protein kinase) and NF-κB pathways are vital components in the efficient allorecognition mechanism of symbionts and pathogens by the coral innate immune system. In the quest for the PRRs in the coral immune system, Toll-like receptors (TLRs) with LRR and TIR domains have been characterized in *Acropora digitifera* [55,56]. The TLR pathway analysis of *Orbicella faveolata* has revealed that coral receptor mediated immune activation is mostly similar to LPS mediated TLR4 pathway activation of mammals which ultimately leads to NF-κB activation (Williams et al., 2017). In order to validate the complete pathway, transcriptome analysis of LPS activated coral tissue was performed. Among genes of 13 pathways, NF-κB signalling related 16 genes were significantly expressed which includes, TLR1- and 2- like proteins; members of the tumor necrosis factor associated receptor (TRAF2, 3, 5) and the adapter protein MYD88. The recombinant coral TLR also cross reacts with human MYD88 which corroborates the proposed TLR 4 mediated antigen recognition pathway in corals.

A complex combination of cytoplasmic TIRs domain i.e., NLR-like proteins with various effector domains have also been characterized in *A. palmata*, *A. millepora*, *A. digitifera*, *O. faveolata* and *Montastraea faveolata* [56,57]. Meta-analysis revealed a larger number of NACHT [NAIP (neuronal apoptosis inhibitory protein), CIITA (MHC class II transcription activator), HET-E (incompatibility locus protein from *Podospira anserina*) and TP1 (telomerase-associated protein)] and related domain nucleotide-binding adaptors shared by APAF-1 (Apoptotic protease activating factor-1), R proteins, and CED-4

(*Caenorhabditis elegans* cell death protein 4) (NB-ARC)-encoding loci. The N-terminal effector domains included the apoptosis-related domains caspase recruitment domain (CARD), death effector domain (DED) and C-terminal repeat domains included LRRs, tetratricopeptide repeats, ankyrin repeats, and WD40 (Tryptophan-aspartate motifs) repeats. The coral NLR receptors are activated by MAMPS (Microbe associated molecular patterns) and DAMPS (Damage associated molecular patterns) and the final process mostly leads to apoptotic cell death. The comparative analysis of NLR repertoires of *A. digitifera* and *O. faveolata* has revealed that domain mediated effector signalling varies in corals. Unlike the former one, adaptor protein containing DED and DD (Death domain) domains has been identified in *O. faveolata* which may expand the scope of NLR-caspase interactions into inflammasome formation [58]. Similar adaptor protein families have been identified in *A. digitifera* containing both DD and CARD domains [57]. A repertoire of TIR-domain-containing proteins has been characterized from the corals Acroporids and Pocilloporids [59].

Mannose-binding lectins have been successfully characterized for the first time in the scleractinian coral *Acropora millepora* and it was named Millelectin. Millelectin was found to be closely related to the vertebrate MBL (Mannan binding lectin), Surfactant protein-A, and the ascidian MBL-like protein through phylogenetic analysis [60]. Also, the existence of complement protein C3-like molecules has been demonstrated in *A. millepora* (C3-Am) [56]. Thus, a primitive complement pathway involving the allorecognition and binding of lectin with C3 followed by direct activation of phagocytosis as seen in invertebrates [61,62] is interesting scope for prospective research studies in the context of coral innate immune system. This understanding will prove to be useful in comprehending the functional relationship between Millelectin and C3-Am in coral innate immunity. Furthermore, TNF receptors (TNFRs) and TNF receptor-associated factors (TRAFs) are also implicated in the thermal stress response via apoptosis [63] and activation of nuclear factor-κB (NF-κB/Nfkb1) and c-Jun N-terminal kinase (JNK/MAPK8) associated immune response cascade. Also, a lectin-like gene, namely Techylectin, has been described from the scleractinian coral *Oculina sp.* [64]. In *A. millepora*, multiple integrins have been identified but their immunological role is yet to be characterized [65,66].

Similar to all invertebrates, corals lack B or T cells mediated

immunity. The “coral probiotic hypothesis” speaks more of exposure mediated tolerance to the environmental and disease related stress. In an experimental challenge with *Vibrio shiloi*, *O. patagonica* was unable to bleach which was previously susceptible to the same bacteria [67,32]. There exists an intricate mechanism by which the attached microbes could be lysed or eliminated from internal organs before it could multiply. Such resistance has also been found in case of type II disease bacterium which are no longer pathogenic to its host coral [68]. Similar resistance mechanism to abiotic stress factors i.e., rise in temperature is being explained by means of the “adaptive bleaching hypothesis” [69]. It has been reported that change in temperature affects the symbiosis of coral holobionts as it is altered and adapted by more heat tolerant organisms. The stressed animal after a period of days could restore more thermal tolerant and disease resistant holobionts which constitutes an adaptive mechanism of corals.

The retrospective analysis of research studies revealed a tremendous number of efforts all over the world in understanding the immune response mechanism of the coral community. Still, there is a lot of scope as several players of immunity found conserved in invertebrates are yet to be characterized in corals. The role of intricate symbiotic association of corals with algae and bacteria in the immune mechanism of corals warrants further elaborate studies.

The role of immune molecules associated with coral immunity and their possible functions were depicted in Fig. 3. Corals rely on their symbiotic relationships with microbial communities for immune defense, influenced by environmental factors. Beneficial bacteria in coral mucus produce antibiotics that protect against pathogens, yet coral health is threatened by several factors like pollution and global warming. Coral immunity involves physical barriers, antimicrobial peptides and immune molecules like heat shock proteins and lectins. They deploy cellular responses like phagocytosis and melanization for wound repair and pathogen defense. Coral innate immunity includes Toll-like receptors (TLRs), NOD-like receptors (NLRs) and lectins, aiding in stress response and pathogen recognition.

Jellyfish immunity

Analogous to the other basal metazoans under the phylum Cnidaria, jellyfish are also characterized by simple body organization with only two tissue layers serving as physical barriers. The simple body organization with lack of adaptive immunity coupled with the constant exposure to a wide array of abiotic and biotic stress agents did not deter these metazoans from evolving alongside the vertebrate community. Undoubtedly, the innate immunity of basal metazoans is highly equipped and more complex than the immune system of their vertebrate counterparts. However, the understanding of the immunological features in basal metazoans like jellyfish is hindered by the practical difficulty in mimicking the same environmental conditions found in the wild. Comprehensive proteomics and transcriptomics studies of the innate immune system in jellyfish are relatively minimal and this provides immense scope for prospective research studies.

Jellyfishes are characterized by soft, fragile and gelatinous body mass. Thus, the physical barriers are inadequate to constitute the first line of defense. Consequently, these metazoans have developed specialized structures called nematocysts that can secrete venom when stimulated by external threats. The sting of a jellyfish can paralyze its prey (feeding) or its predator (self-defense). The body of a jellyfish comprises an outer epidermis, a middle gelatinous mesoglea and an inner gastrodermis. The epidermis is covered by the mucocytes which secrete mucus as a defense strategy. The venom secreted by the nematocysts can also act as an effective defense tactic against potential predators. With reference to the presence of a cocktail of bioactive compounds present in the mucus of other animals, a comprehensive study in understanding the constitution of the mucus barrier in box jellyfish revealed the presence of several hemolytic proteins. Hemolytic toxins, including CaTX-A (43 kDa) from *Carybdea alata*, CrTXs (43–45

kDa) from *Carybdea rastoni*, and CqTX-A (44 kDa) from *Chiropsalmus quadrigatus*, were identified in nematocysts [70,71]. Similarly, transcriptomics and venom proteomics studies in the jellyfish *Chironex fleckeri* identified more than 170 potential toxin proteins, including metalloproteinases, an alpha-macroglobulin domain containing protein, two CRISP proteins, a turriptide-like protease inhibitor, and particularly, nine novel members of a taxonomically restricted family of cnidarian pore-forming toxins [72]. Moreover, 174 potential toxic proteins including phospholipase A2, zinc metalloproteinase-disintegrin agkistin, serine protease inhibitor, plancitoxin-1, alpha-latrocrustotoxin-Lt1a, etc. were identified in the jellyfish *Cyanea nozakii* [73,74]. In the jellyfish *Aurelia coerulea*, 21 self-protective proteins of mucus, such as metalloproteinases, serine proteinase inhibitors, superoxide dismutase's (SODs) and complements were scanned against potential external invasion [74].

Potential symbiotic type relationship may be possible since the microbiome analysis revealed the presence of microbiota associated with moon jellyfish which was distinct from the bacterial community in surrounding seawater and was also found to vary across different body parts. Betaproteobacteria (Burkholderia, Cupriavidus and Achromobacter) was dominant in the gastral cavity of medusa, while Alphaproteobacteria (Phaeobacter, Ruegeria) and Gammaproteobacteria (Stenotrophomonas, Alteromonas, Pseudoalteromonas and Vibrio) prevailed on ‘outer’ body parts [75]. The study was not able to conclude whether the bacterial association was the result of opportunistic invasion or ‘true’ symbiosis. Since the microbiome composition is different from that of the seawater, there is a possibility of regulated selection by the host. The microbiome analysis of *Aurelia aurita* also revealed the life stage-specific distinction in the microbial colonization [76]. Detailed investigational study is warranted to comprehend the relationship between the microbiota and the host.

As an aquatic invertebrate that is in constant direct contact with a microbe-laden environment with no adaptive immunity, the innate immunity of jellyfish is equipped with potent antimicrobial peptides as the significant effectors of immunity. In support of this proposition, a novel AMP called aurelin has been identified in the mesoglea of scyphoid jellyfish *Aurelia aurita* with potent antimicrobial effects against Gram-positive and Gram-negative bacteria [77]. A proteinaceous antimicrobial material JAP-1 (Jellyfish antimicrobial protein) has been purified from the whole-body extracts of *Nemopilema nomurai*. JAP-1 exhibited significant antimicrobial activity against *Escherichia coli* D31, *Bacillus subtilis*, *Streptococcus iniae* and *Candida albicans* [78]. Since jellyfish undergoes external fertilization in the microbe-laden waters, the eggs are also equipped with certain defense factors against microbial colonization. The very first case of antimicrobial activity of jellyfish eggs was identified in the oocyte lysate of *Rhizostoma pulmo* which exhibited an antibacterial lysozyme-like activity [79].

As a self-repair strategy against adverse physical trauma, the jellyfish is equipped with the ability of reorganization and symmetrization which is different from regeneration. Symmetrization has been observed in the following jellyfish: *Aurelia aurita*, *Chrysaora pacifica*, *Mastigias* sp., and *Cotylorhiza tuberculata*. The study inferred that this distinct mechanism is driven by sheer mechanical forces and highlighted the need for understanding the evolutionary trajectory with respect to regeneration [80]. Retrospective analysis of research works in profiling the immune components and the mechanism in jellyfish revealed comparatively limited studies and this aspect offers immense scope for prospective research endeavors.

Overall, Jellyfish, similar to other basal metazoans in the phylum Cnidaria, have simple body structures lacking adaptive immunity. They rely on physical barriers such as nematocysts and mucus, which contain venomous toxins like hemolytic proteins for defense. Microbiome studies suggest a potential symbiotic relationship with distinct bacterial communities across different body parts. Their innate immunity includes potent antimicrobial peptides like aurelin and JAP-1, aiding in defense against pathogens despite constant exposure to microbial

environments. Jellyfish also exhibit unique self-repair capabilities through symmetrization. Despite limited research, their immune system complexity presents significant opportunities for future investigation.

Sea anemone immunity

The immune gene repertoire found the basal Cnidarians reflect an ancestral and primitive model of the vertebrate immune system. Though certain similarity of innate immune repertoire may be found with respect to the vertebrates, the phylogenetic studies reflect a huge extent of gene loss associated with the divergence of the lineage. The diverse repertoire of innate immune genes observed in these basal metazoans contributes to the complexity of the immune response in these primitive invertebrates.

Similar to their Cnidarian counterparts, the sea anemones have also developed an intricate network of innate immune genes that work together to compensate for the lack of adaptive immunity. The whole genome profile and expressed sequence tag analysis of the non-symbiotic sea anemone *Nematostella vectensis* and the coral *Acropora millepora* revealed a total of 16 571 non-redundant ESTs (10 550 from *Nematostella* and 6021 from *Acropora*) and 12 547 predicted peptides (7484 from *Nematostella* and 5063 from *Acropora*) across the two species [81]. The study highlighted the ancestral basis of genetic complexity of higher animals. Notably, the isolation of twelve Wnt genes from *Nematostella vectensis* has been reported with certain similarity to that of advanced metazoans [82]. Allogeneic and syngeneic recognition by the anemone is mediated by a specific cellular recognition system as evident in an aggregating sea anemone, *Anthopleura elegantissima*. When threatened, peculiar surface protrusions (acrorhagi) are used for stinging with nematocysts. Nematocysts are also found in the oral tentacles and these structures respond to the threat through physical contact, implying the presence of an array of allorecognition receptors on the ectoderm. These receptors play an important role in distinguishing between allogeneic and syngeneic entities. Similar response maybe seen in *Actinia equina*, *Anthopleura artemisia*, *Anthopleura krebsi*, and *Metridium senile* but not in *Anthopleura xanthogrammica* [83].

The sea anemone *Anthopleura xanthogrammica* are unresponsive towards the adjacent syngeneic clones but elicit rejection of heterogeneous clones [84]. Screening of the EST and genomic information revealed the existence of five TIR proteins: an MyD88 homolog (NvMyD88), a protein (NvTLR-1), three interleukin 1 receptors NvIL-1R1, NvIL-1R2 and NvIL-1R3. The comprehensive analysis of the immune gene repertoire of *Nematostella* by Miller et al. [56], reveals the presence of a wide array of immune genes related to the TLR pathway [LBP (Lipopolysaccharide binding protein), TLR, MyD88, IRAK, TRAF6, TAK1, I κ B (inhibitor of nuclear factor- κ B kinase), NF- κ B], ILR pathway [TRIF [TIR (Toll/ interleukin-1 receptor/ resistance protein) domain-containing adaptor inducing interferon- β], TRAM (TRIF-related adapter molecule), IRF3], ECSIT pathway (Evolutionarily conserved signalling intermediate in Toll pathways) MEKK1/MAP3K (Mitogen-activated protein kinase 1), MKKs (Mitogen-activated protein kinase), JNK, p38, AP1, ATF (Activating transcription factor), complement system proteins (C3/A2M related) and membrane attack complex/perforin domain MAC/PF domain-containing proteins (Tx60-A, MPEG). Activation of NF- κ B signalling in the Nv-TLR pathway against *V. coralliilyticus* and flagellin has been characterized in the sea anemone, *N. vectensis*. Further analysis states that in adult anemones, body column and nematosome cnidocytes that express Nv-TLR also often express Nv-NF- κ B, but tentacle cnidocytes almost entirely express only Nv-TLR, and nearly half of body column cnidocytes express Nv-NF- κ B but not Nv-TLR. The study has postulated the possibility of both Nv-NF- κ B-independent (in tentacle cnidocytes) and Nv-NF- κ B-dependent (in nematosome and body column cnidocytes) pathways in adult animals. Nv-TLR has a NF- κ B-independent role in early embryonic development [85]. NF- κ B transcription factor expression (Ap-NF- κ B) in *Exaiptasia pallida* has been correlated with regulation of symbiosis [86]. The cGAS-cGAMP-STING

pathway is an effective antiviral immune cascade of mammals resulting in the type I interferon response. Though interferons are absent in invertebrates, the STING (Stimulator of interferon genes) pathway has been reported in the anemone *Nematostella vectensis* and the exact role is unclear. cGAMP, an activator of the STING pathway, has been reported to induce both antiviral and antibacterial response [87]. Several constituents of complement system like the factor B genes (Ap_Bf-1, Ap_Bf-2a, and Ap_Bf-2b) and one MASP (Mannose-binding lectin associated serine proteases) gene (Ap_MASP) have been identified in *A. pallida* [88]. Partial sequences of C3, two factor B (Bf), and one MASP genes have been reported in *N. vectensis* [89]. The unexpected existence of diverse innate immune genes reflects the efficiency of the innate immune system to compensate for the adaptive immunity. The gene loss episodes did not deter the activity of the innate immune response in these basal metazoans.

Scavenger receptors (SRs) are constituents of the allorecognition mechanism and are able to recognize several microbial ligands and produce a phagocytic response. Thirty six SR genes have been identified in the genomic profile of *A. pallida* and other Cnidarians which is higher compared to humans (16SRs) [90]. Thrombospondin-type-1 repeat (TSR) domain proteins which have been identified in *A. pallida* are involved in allorecognition and subsequent selection of symbionts. They contribute to the symbiosis specificity by promoting selective colonization of symbionts [91]. An important cellular response against thermal stress constitutes the upregulation of nitric oxide (NO) synthesis. Cnidarian bleaching in the symbiotic *A. pallida* is mediated by the nitric oxide synthesis by the host resulting in the increased oxidative load with the subsequent cytotoxic effect and bleaching of symbionts [92]. A caspase-mediated, NO-dependent pathway against thermal stress has been identified in *A. pulchella* (Hawkins et al., 2013). In an attempt to study the mechanism of symbiont accommodation in hosts, Chen et al. [93], characterized the existence and activity of Rab7 protein (ApRab7) in *A. pulchella*. The interesting variation in the expression pattern of Rab7 in response to live algal symbionts and the heat-killed or photosynthesis-impaired symbionts helps to comprehend the specificity of the symbiont colonization process. The live and photosynthesis-competent zooxanthellae excluded the expression of ApRab7 on the phagosomes, and therefore, persist within the host. The heat-killed and photosynthetically incompetent symbionts displayed the ApRab7 expression, thus negatively affecting the establishment of symbiosis [93]. A novel ancient caspase (acasp) and a Bcl-2 family member (abhp) containing two Bcl-2 homology (BH) domains have been identified in *A. pallida*, highlighting the existence of an ancient proto-type of apoptosis [94].

A critical look into the retrospective analysis of the immune gene profile of sea anemones revealed a significant level of conservation and an unexpected degree of genetic complexity in spite of/because of ancestral gene loss during the course of evolution. The remarkable similarity in the immune gene profile of sea anemone (primitive invertebrates) with respect to the distant evolutionary clades (higher vertebrates) helps to comprehend the transitional trajectory of the innate immunity along the course of evolution. Briefly, Sea anemones possess a diverse array of innate immune genes that compensate for the lack of adaptive immunity. Studies on few species reveal a complex immune gene repertoire including TLRs, NF- κ B pathways, complement proteins and scavenger receptors. They mediate allogeneic recognition through surface receptors and contribute to symbiosis specificity. Additionally, mechanisms like nitric oxide synthesis and apoptosis play roles in responses to thermal stress and symbiont accommodation. Despite evolutionary gene losses, sea anemones exhibit surprising immune genetic complexity, offering insights into the evolution of innate immunity across distant taxa.

Echinodermata

Phylum echinodermata comprises benthic invertebrates which plays

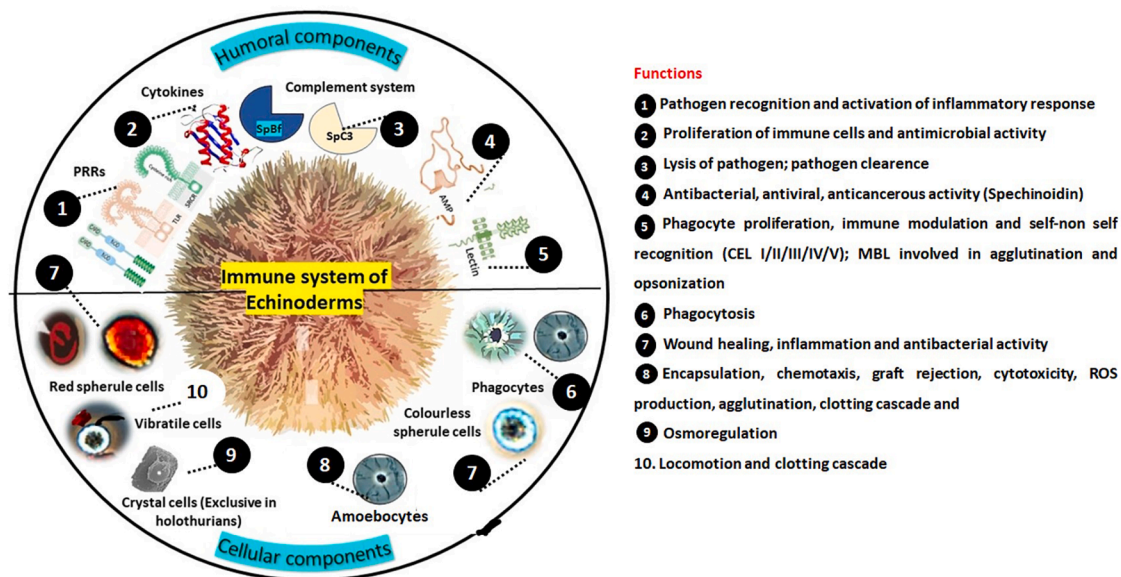


Fig. 4. Immune molecules reported in the phylum Echinodermata and their possible functions.

a key role in many benthic ecosystems in the ocean. Echinoderm is a Greek word which means “Spiny skin”. Sea cucumbers (holothurians), sea urchin (echinoids) and sea stars (crinoids, asteroids and ophiuroids) are the well-known echinoderm species. These animals were the first, used by Metchnikoff who demonstrated the encapsulation of foreign particles by immune cells of an invertebrate. Consequent to that, experiments on allograft rejection determined that they can differentiate between self and non self. Several studies have been conducted on the characterization of the immune system of echinoderms were carried out worldwide. The immune system of echinoderms is entirely innate in nature that is effective for host protection without adaptive capabilities. However, it has been found to be highly sophisticated and complex. Humoral innate responses such as antimicrobial peptides, lectins, complement like factors, agglutinins, cytokines, lysins, prophenoloxidase and cellular innate responses i.e., clotting, encapsulation, clumping, opsonization and phagocytosis were reported in echinoderms.

Sea urchin

Sea urchins are spiny, globular, sedentary marine invertebrates with radial symmetry which are exploited for their valuable bioactive compounds. They are also used as a model system for human immunity related studies [95]. Immunity of sea urchin totally relies on the innate immune system to protect themselves against the pathogens. Studies on the immune system of sea urchin have commenced quite earlier. Self and non self-recognition of these animals after the injection of sheep RBC was reported by Ito et al. [96]. He recorded the increased production of reactive oxygen intermediates after the injection of sheep RBC. It has the capacity to remove 90–99 % of the foreign particle inoculated to purple sea urchin reported by Yui and Bayne, [97].

Humoral components of the immune system include i.e., PRRs (SpTLR, SpNLR, SpSRCR), complement system, lysozyme, antimicrobial peptides, lysins and C-type lectins. SRCR receptor transcripts are highly expressed in coelomocytes in echinoids, however, it is altered after bacterial, fungal infection or during injury. SpTLR structure is similar to other TLR molecules with cytoplasmic TIR domain which facilitates signalling, and a LRR ectodomain [95]. TLR signalling pathway associated genes, homologous to vertebrate NFkB i.e., SpNFkB (transcription activator), SpGATAC (Repressor) and SpRunt (Embryos) are reported in sea urchin, their expression profiles are higher after 6–12 h of bacterial challenge. Complements of echinoids are homologous to the vertebrate alternate complement pathway. The components such as SpC3 and SpBf

have shown significant similarity between the sequences of vertebrate C3 and Factor B complement molecules. Apart from these the gene sequence of C1q, C2, C3, C4, C5, mannose binding lectin, ficolin, CD59 and thioester containing protein are encoded in the genome of purple sea urchin [98]. It clearly indicates the complement system of vertebrates has evolved from the simplified sea urchin complement system which underwent duplication and increased complement components during the evolution [99]. Lysozyme enzymes with antibacterial activity have also been observed in the red spherule cells of the sea urchin. Several molecules with anti-microbial properties are reported in sea urchins. Among them Spstrongylocin is an antimicrobial peptide which is effective against a broad range of gram positive and gram-negative bacteria pathogens. SpEchinoidin (homologous to C-type lectin) along with the wide range of lectin molecules are reported in sea urchins following immune challenges. Based on the amino acid analysis, SpEchinoidin is a homodimer and the CRD (Carbohydrate Recognition Domain) of SpEchinoidin binds with galactose and its derivatives [99]. There are several proteins involved in the cytoskeletal modification (profilin, a and b-tubulin, gelsolin, dynein heavy chain, kinesin light chain, fascin, actin and thymosin-b) of coelomocytes are reported. Higher expression of these protein molecules during LPS challenge suggest that extensive cytoskeletal remodeling is required after the immune challenge in echinoids. Furthermore, genes encoded for cytokines such as IL-17 and MIF are also found in sea urchin. An exclusive gene family i.e., Sp185/333 (renamed as Sptrf protein gene) transcripts which are upregulated after the immune challenge is reported in sea urchin. It exhibits conformational elasticity therefore can lead to flexible binding with a broad range of foreign molecules and pathogens. Another fascinating discovery is the identification of SpRAG (Recombination activating gene in mammals) genes from sea urchin, sea star and brittle star. These findings imply that an ancient RAG1/RAG2 gene pair was present in the genome of the last common deuterostome ancestor, prior to the development of the adaptive immune system in jawed vertebrates. Besides, their exact roles in echinoderm immunity are still unknown; however, it is possible that they have a role in gene rearrangements.

Cellular factors including phagocytosis, encapsulation, clotting, inflammation, cytotoxic and cytolytic responses which are regulated by the conserved immune cells in echinoids, called coelomocytes. Based on morphological characteristics of coelomocytes, at least 6 types of cells are described in echinoderms. However, all the six cells are not discovered in all classes and species. Phagocytes (also known as leukocytes), vibratile cells, spherule cells (also known as granulocytes,

amoebocytes, or morula cells), hemocytes and progenitor cells are the coelomocytes reported in echinoids [100]. Phagocytes are the most common coelomocytes, have the capacity to alter shape from a petaloid/bladder to a filopodial form, which is important for phagocytosis, encapsulation, CF clotting and syncytia formation. It is also reported to produce antimicrobial peptides and is involved in cytotoxic and cytolytic responses. Opsonization aids to enhance phagocytosis and encapsulation in echinoid immunity [101]. Major opsonin molecules reported in sea urchin are SRCR, lectins and proteins of the complement system. Spherule cells are ovoid in shape and contain granules with mucopolysaccharide and protein. It is involved in the cellular responses such as inflammation, encapsulation, wound healing, cytotoxic activity and also exhibits anti-bacterial against several pathogens [102]. Vibratile cells are motile and identified with their single long flagellum. They are reported to be involved in clotting cascade and help to maintain the homeostasis [103]. Large coelomocytes with hemoglobin and involved in oxygen transport are recognized as hemocytes [104]. Similar to small lymphocytes of vertebrates, a cell population found in echinoids are called progenitor cells [101]. Clotting cascade is mediated by coelomocytes and coelomic fluid (CF). Amaasin, a CF protein, is considered as the prime mediator of clotting in echinoids [105]. During cellular clot formation, a domain called olfactomedin in the amaasin protein facilitates the intercellular adhesion of coelomocytes. It is essential to understand the immune system of echinoids because they are present at the base of deuterostome lineage in the phylogeny as a sister phylum to the chordates. The studies on the functioning of sea urchin immune system and comparing it with vertebrate and other invertebrate immune systems will be helpful in understanding the historical evolution of immune responses.

Overall immune responses produced by sea urchin and the immune molecules responsible for holothurian immunity and their possible functions were depicted in Fig. 4. Sea urchins rely exclusively on their innate immune system for defense against pathogens. Their immune repertoire includes pattern recognition receptors (PRRs), complement proteins, lysozymes, antimicrobial peptides and lectins. Cellular immunity is mediated by coelomocytes, including phagocytes, spherule cells, vibratile cells, hemocytes and progenitor cells, which perform functions like phagocytosis, encapsulation, clotting and cytotoxic responses. Unique immune genes like Sp185/333 and SpRAG suggest ancient immune mechanisms. Studying sea urchin immunity informs our understanding of immune system evolution across deuterostomes.

Sea cucumbers

Sea cucumbers are commercially important echinoderms and also well known as a model system to study regeneration. *Holothuria scabra*, *Apostichopus japonicus* and *H. Spinifera* [106] are the major species of commercial holothurians cultivated worldwide. The innate immune system of sea cucumbers has the following characteristics; capacity to differentiate self and non self, effective wound healing ability, and coelomocyte cellular functions. The non-self-recognition capacity of sea cucumbers was assessed by [107]. They noticed the clearance of sheep red blood cells within 8 days of injection from the sea cucumber *Holothuroidea polii*. Their findings corroborate with the emergence of brown bodies in the coelomic cavity [107]. The aggregates of RBCs surrounded by coelomocytes found inside the brown bodies, indicated the activation of phenoloxidase and melanin production. Apart from humoral and cellular defense mechanisms, various unique features and immune molecules are described in this incredible animal. Lectins, lysins, pathogen recognizing receptors, antimicrobial peptides, enzyme inhibitors and complement like systems are the humoral components reported in holothurians immunity.

Lectins are reported to be involved in modulation of inflammation, self and non-self-differentiation and other autoreactive processes [108]. CEL (*Cucumaria echinata* lectin) I/II, CEL-III, CEL-IV from *Cucumaria echinata*, ii) SPL (*Stichopus japonicus* lectin) -I, SPL-II, SJL-I from

Stichopus japonicus and iii) MBL-C (Mannon binding lectin from *Cucumaria* sp) from *Cucumaria japonica* are the lectins reported from sea cucumbers. Function of CELs (Mitogenic lectin) is to aid the proliferation of immune cells while MBL-C contributes in opsonization and agglutination but not in formation of membrane attack complex (MAC). Among the holothuroid lectins, CEL-III is the most well-studied hemolytic lectin. It binds to Gal/Gal NAc (Galactose/Galactose N-Acetyl glucosamine) and undertakes a conformational shift that serves as a complement to cause hemolysis in the target cell. This lectin binds to the specific ligand (Gal/Gal NAc), in the presence of Ca²⁺, at the non-reducing ends and CEL-III oligomerizes to form pores on the membrane, resulting in membrane osmotic rupture [109]. As part of the innate immune response, anti-microbial peptides provide the first-line protection against invading pathogens. The AMP compounds identified from *Holothuria tubulosa* [110] were Holothuroidin 1(H1) and Holothuroidin 2 (H2), whilst anti-microbial peptides A1, A2, and A3 were recovered from *Apostichopus japonicus* [111]. These compounds were shown to limit bacterial and cancer cell multiplication by carpet killing mechanisms. Toll-like receptor genes i.e., AjTLR3 and AjToll and TLR pathway downstream molecules i.e., AjMyD88, AjTRAF6, Aj-ikb, Aj-105, Aj-p50 and Aj-rel identified from *A. japonicus*, are confirming the existence of the TLR pathway in sea cucumbers. However, further research investigations are required to discover the remaining components to obtain the complete TLR pathway. Wang et al. [112] observed an intense increase in the expression of TLR genes in the coelomocytes of sea cucumbers after the exposure of peptidoglycan. The complement system aims to enhance pathogen clearance in the body. AjC3 and AjC4 are the complement analogues with haemolytic activity detected in the coelom of *A. japonicus*. Consequently, AjC3-2, a new C3 isoform was discovered in the sea cucumber (*A. japonicus*) ([113]. In the coelom of *Holothuria leucospilota*, Xing and Chia [114] identified a 20 kD opsonin-like protein that promotes phagocytosis. AjC3-2 and AjC3 have been reported to perform like opsonin in various echinoderm species [115]. Enzyme inhibitors reported in holothurians are lysozyme and phenol oxidase. Lysozymes are antibacterial agents and play a key role in phagocytosis in sea cucumbers. SjLys (*Stichopus japonicus* lysozyme) identified from Japanese sea cucumbers. Its amino acid sequence is reported to be similar to i-type lysozymes. Glycosidase and isopeptidase are the two active sites in lysozyme due to which it is regarded as a chimerically bifunctional enzyme. SjLys in its native as well as heat treated form displayed a wide range of antibacterial activity against gram negative bacterial pathogens such as *Vibrio parahaemolyticus* and *Pseudomonas aeruginosa* [116]. He also mentioned that owing to its enzymatic and non-enzymatic antibacterial activity, it is evident that it can act like 'magic bullets' for the pathogens. Phenol oxidase (PO) is observed in sea cucumbers. It converts the phenol substrates into unstable quinones which are further transformed into melanin via a non-enzymatic pathway. AjPO1, AjPO2 and AjPO3 are the three phenol oxidase enzymes identified from Japanese sea cucumbers. All of them are laccase type phenol oxidase which catalyse the oxidation of methoxy phenols, polyphenols, diamines and several other compounds. Presence of Ca²⁺, Mg²⁺ and Mn²⁺ strongly increases the activity of propenol oxidase enzymes.

Cellular immunity in holothurians is assisted by coelomocytes. They are present in the coelomic cavity and can invade various tissues. Research investigations on coelomocytes found that exposure of PAMPs to sea cucumbers didn't alter the total coelomocyte number whereas other type of coelomocyte such as lymphocytes were reduced and phagocytes were increased [117]. The subpopulations of coelomocytes are amoebocytes, vibratile cells, crystal cells, spherule cells and lymphocytes reported in sea cucumbers. Amoebocytes are termed as phagocytes and play a major role in several cellular responses such as encapsulation, chemotaxis, graft rejection, cytotoxicity, ROS production, agglutination and clotting cascade. Lymphocytes are the most common coelomocytes observed and the progenitor of all other types of coelomocytes. Spherule cells are identified by the presence of vesicles in

Table 1
Components of Non bilaterian invertebrate immune system and their functions.

Phylum	Symbiotic microbial defense	Humoral factors	Immune cells	Cellular functions	References
Porifera sponges	Bioactive metabolites exhibiting antibacterial and antifungal properties; Chemical defense by calyculin compounds; Passive prokaryotic defence (CRISPR)	PRRs (TLR, NLR, SRCR, CAS c, IRAK); Lectins, AMPs, Interferons	Pinacocytes, exo-pinacocytes, multi-granulated gray cells	Physical barrier phagocytosis inflammation apoptosis, allorecognition	[25]
Ctenophora Sea walnut	ND	Lectins Complement	Mesenchymal cells of mesoglea, amoebocytes	Physical barrier, phagocytosis, extracellular trap, inflammation, allorecognition, ETosis	[28,30]
Cnidaria Corals	Wound activated chemical defense; antimicrobial property	Thermal stress tolerance (HSPs, antioxidants, apoptosis regulator; cell to cell adhesion and calcium ion homeostasis); AMP (Damicornin); ProPO system; PRRs (TLR & NLR pathways); Lectins (MBL); Complement system.	Sclerite layer cells Granular cells Amoebocytes	Physical barrier, coagulation cascade, phagocytosis, inflammation, melanization, allorecognition	[42,43]
Jellyfish	Life stage-specific distinction in the microbial colonisation is observed. But, the actual mechanism is not yet determined	Mucus (hemolytic proteins; metalloproteinases, alpha-macroglobulin domain containing protein, two CRISP proteins and pore forming toxins); AMPs (aurelin);	Specialized defence structures called nematocysts produce venom; mucocytes; mesogleal cells	Physical barrier, Allorecognition, reorganisation and symmetrization	[72–74, 77]
Sea anemone	External defense; antibacterial and antifungal; homeostasis	PRRs (TLR, SR, ILR, ECSIT pathway); complement system (C3/A2M related & MAC/perforin protein); STING pathway (antiviral); lectin/glycan interaction;	Nematocysts	Allorecognition, upregulation of nitric oxide synthase, apoptosis	[56,128]
Echinoderms Sea urchin	Homeostasis, helps in pathogen removal	PRRs (SpTLR, SpNLR, SpSRCR); complement system; lysozyme; AMPs (Spstrongylocin) and C-type lectins (SpEchinoidin)	Coelomocytes Phagocytes, vibratile cells, spherule cells, hemocytes and progenitor cells	Allorecognition, phagocytosis, encapsulation, clotting, inflammation, cytotoxic and cytolytic responses	[99,101]
Sea cucumber	Antibacterial and antifungal properties	PRRs (TLR); lectins (CEL I/II/III), AMPs (Holothuridin 1), enzyme inhibitors and complement like system	Coelomocytes, lymphocyte, vibratile cells, spherule cells, amoebocytes, crystal cells	Phagocytosis, encapsulation, brown body formation, cellular aggregate formation	[101]
Sea star	Antibacterial and antifungal properties	PRRs; complement system; lysozyme; AMPs, lysins, saponin and saponin like compounds, HSP 70 and ROS production, cytokines (IL-q & IL-6) and C-type lectins	Coelomocytes i.e., phagocytes and colorless spherule cells	Phagocytosis, arm regeneration, microbial clearance and clumping	[120,121]

their cytoplasm. They are involved in wound healing, inflammation and antibacterial activity. Vibratile cells have a projecting flagellum used for locomotion. It is majorly involved in the movement of CF and clotting cascade. An exclusive coelomocyte in sea cucumbers is crystal cells which are believed to be related with osmoregulation. Apart from the cellular responses such as phagocytosis and encapsulation, cellular aggregate formation (clotting) and brown body formation are the two important mechanisms facilitated by coelomocytes. The cellular aggregates formed after the injury can crawl and translocate to other places by which the injured site will be closed [101]. Pigmented cell aggregates with unwanted material are called brown bodies. Spherulocyte and amoebocyte form cellular aggregations around the invaded foreign material and surround them by a fibrillar material. These aggregated particles are fragmented with the help of lysozymes and transferred to cloaca where it will be eliminated from the system [118]. Xue et al. [119] tabulated the immune related genes and their tissue distribution in sea cucumbers primarily detected by expressed sequence tag (EST) and proteins and their possible functions.

Similar to other echinoderms, sea cucumbers rely on their innate immune system for defense, characterized by self-nonself recognition, wound healing and diverse coelomocyte functions. Their humoral immunity includes lectins, antimicrobial peptides, enzyme inhibitors like lysozyme and phenol oxidase and complement-like systems. Toll-like receptors and downstream signalling molecules are identified, enhancing understanding of their immune pathways. Coelomocytes play pivotal roles in cellular immunity, encompassing phagocytosis, encapsulation and clotting. Unique cell types of echinoderms include amoebocytes, spherule cells, vibratile cells and crystal cells, involved in various immune responses and tissue repair mechanisms.

Starfishes

Sea stars in the phylum echinodermata includes asteroids (sea stars), sea lilies and feather stars (crinoids), ophiuroids (brittle stars). They are well known for regeneration of their arms. The immune system of starfish is almost similar with other echinoderms. Only the differences existing in the starfish immune system as well as the variations observed between different classes of starfish are discussed below.

Lysins, also known as hemolysins, are lytic chemicals found in sea stars which form transmembrane pores on the pathogens [120]. Complement component C3 is reported in sea stars and the increased expression is observed during LPS challenge [121]. Anti-microbial agents i.e., saponin and saponin like compounds are used to maintain antiseptic environment in starfishes. Asteroids and ophiuroids release steroidal glycosides that show anti-fungal properties and exhibit toxicity against brine shrimps. Hsp 70 and ROS production are noticed in the sea stars when they are exposed to heavy metals such as cadmium and polychlorinated phenols [122–124]. Apart from these cytokines such as IL-1 and IL-6 are reported in sea stars. Lectin molecules are also identified from the hemolymph of starfish [125].

Asteroids are reported to have only two types of coelomocytes i.e., phagocytes and colorless spherule cells. Due to the ability of the phagocytes to rapidly differentiate into several functional cell types, efficient microbial clearance is possible in sea stars. Ophiuroids and crinoids contain phagocytes, colorless spherule cells, whereas crystal and vibratile cells found in ophiuroids and red spherule cells are observed in crinoids. Phagocytes and red spherule cells seem to be involved in arm regeneration in star fishes [126,104]. These results indicate that increased coelomocyte count and the expression of classic stress molecules such as Hsp 70 plays an important role in the early

Table 2
Components of bilaterian invertebrate immune system and their functions.

Phylum	Symbiotic microbial defense	Humoral factors	Immune cells	Cellular functions	References
Mollusca Cephalopods and bivalves	Antimicrobial property	Opsonin like compound, lectins (agglutinin), PRRs (TLR), Thioester-containing proteins (TEP), C-type lectin and down syndrome cell adhesion molecule (DSCAM), enzyme inhibitors (lysozyme, chitinase and cathepsins)	Mucocytes, hyaline cells, blastlike cells and granular cells	Phagocytosis, encapsulation, melanization, cell adhesion, cell aggregation, secondary immune memory, opsonization	[131,132]
Arthropoda Shrimp, crab and lobsters	Anti-microbial, antifungal property, homeostasis	PRRs (TLR, NLR, RLR), cytokines, C type lectins, AMPs (Penaeidin), PPO system, immunoglobulin like proteins, acute phase proteins and lysozyme	Hyaline cells, granular and semi granular cells	Phagocytosis, apoptosis, encapsulation, clotting cascade and nodule formation	[133,134]

repair of tissue injury and regeneration. Increased phagocytic activity has been found when the animals are exposed to heavy metals in sea stars [123]. In sea stars, clumping of multicellular aggregates is considered as an alternative reaction to phagocytosis [127].

In general, Sea stars exhibit a robust innate immune system with similarities across echinoderms. They utilize lysins, complement components like C3, antimicrobial agents such as saponins and produce Hsp 70 and ROS in response to stressors. Cytokines like IL-1 and IL-6, as well as lectins, contribute to their immune defense. Coelomocytes in sea stars vary by class; asteroids have phagocytes and colorless spherule cells, while ophiuroids and crinoids possess additional cell types like crystal cells and red spherule cells. These immune mechanisms support their remarkable abilities in regeneration and defense against pathogens and environmental stressors. Overall, the major components involved in the immune system of non-bilaterian invertebrate were tabulated in Table. 1.

Immunity of bilaterian

As the name suggests, they have a bilateral symmetric body and quit advanced body features. The comparative genomics has suggested almost 157 specific genes of bilaterian, involved in Nodal pathway, a key regulator of mesoderm development and left-right axis specification; components for nervous system development, including a suite of G-protein-coupled receptors, having important role in evolution [129]. Apart from advancement in excretory, digestive and nervous systems, the immune mechanism also carries novel proteins, peptides, receptors and immune pathways. The intraspecific variations in immune pathways suggests that immunity has a great role in macroevolutionary events [130]. Among many animals, annelids and arthropods immune systems are selected here for discussion. The major components involved in the immune system of bilaterian invertebrate was tabulated in Table. 2.

Mollusca

The maximum likelihood analysis of 18S rRNA gene sequence has clarified that Annelida and Mollusca are sister groups, which together comprise the Eutrochozoa, a group having similar larval trochophore stage [135]. Being the second largest animal group, phylum mollusca includes three sub groups i.e., gastropods, bivalve and cephalopods. The respective immunity of three subgroups is mentioned here.

Most of the immune defense related studies have been focused on land and freshwater snails whereas marine gastropods are seldom used for this kind of investigation. The ciliated epithelium carrying mucocytes, comprise a first line of defense to capture and shade the microbes. The mucus glycoprotein and polysaccharides impart chemical defense against microbes as mentioned in case of giant African snail [136]. Studies on humoral and cellular immune defense of Sea hare *Aplysia californica*, and the giant keyhole limpet *Megathura crenulata* has imparted that plasma carries agglutinin and opsonin like compound which facilitates phagocytosis [137]. There are helical proteins present in plasma which may be useful for controlling blood loss at the site of

injury. There is a single type of hemocytes suspended in the blood which carries lysosomes. Upon antigen recognition these cells are able to migrate and phagocytose the antigen. The cells are able to produce SOD and reactive oxygen for deactivating the microbes. However, there is no PPO enzymes produced in these cells. In contrast the disk abalone, *Haliotis discus* and the spiny top shell, *Turbo cornutus* carries two and four types of hemocytes as investigated by Donaghy et al. [138]. The *H. discus* carries blast like cells and hyaline cells whereas *T. cornutus* carries two types of hyaline cells, granular cells and blastlike cells. The function is similar to other gastropod hemocytes but the degree of phagocytosis may vary between species. The blast cells and hyalocytes carry lysosomes but only later one can perform phagocytosis. The cells are able to produce ROS and NOS. More information related to defense organs and TLR receptors are lacking from marine gastropods.

In addition to the internal humoral and cellular defense, bivalves represent extrapallial immune effector compounds [139]. A large portion of the bivalve mantle cavity is exposed to the open water which encounters many microbes. The extrapallial cavity fluid carries a lot of hemocytes, which have bidirectional movement and provides first line phagocytic defense against microbes. The pallial hemocytes are more phagocytic than circulating cells and hence implies to represent unique surface receptors [140]. The extrapallial mucosal secretion also carries lysozyme and proteases for neutralizing invading pathogens. The lectin like agglutinins is present in the mucous which helps in antigen presentation in bivalves. The circulating hemocytes are major players in bivalve defense. The microscopic analysis of Sydney rock oyster has revealed three major hemocytes i.e., hemoblast like cells, hyalinocytes and granulocytes [141]. All of them are phagocytic but granulocytes are more phagocytic than others. The granulocytes carry phenoloxidase, ROS and nitrous oxide, peroxidase and melanin for inactivating microbes. The cells phagocytose the antigens and larger size antigens are encapsulated by forming thin layers by the hemocytes. The hyaline cells are mostly phagocytic but least participate in encapsulation. The inner layer of melanin deposition is mostly formed by granulocytes. Around 27 TLRs have been discovered in mussels which are mostly hemocyte surface receptors. Other than circulating hemocytes, the pallial organs also express TLRs and associated adaptor molecules i.e., six MyD88s and two SARMs in *Crassostrea virginica*. The plasma also carries defense compounds i.e., defensin, lysozyme, galectin, and protease inhibitors which seems to inactivate or inhibit the microbes from entry [142]. Bivalves undergo a secondary response to infection by having immune memory against microbes which is mediated by few protein receptors i. e., Thioester-containing proteins (TEP), C-type lectin and Down syndrome cell adhesion molecule (DSCAM). The DSCAM belongs to an immunoglobulin superfamily which don't have functional role in gastropods but in bivalves (*Mytilus edulis* and *C. gigas*) it imparts similar function as in case of crustaceans i.e. cell adhesion and hemocyte aggregation [143]. However, there is no isoform diversity as in case of crustaceans. Such immune priming is also performed by TEP (similar to complement protein in other animals) and C-type lectin molecules in bivalves [112,144].

Cephalopods are an advanced molluscan variety which lacks an external shell and includes mostly Octopus, squids, cuttlefish and

nautilus. The humoral and cellular defense mechanism is almost similar to other molluscs. There is a hematopoietic organ called “white body” present behind the orbital pits figured out of numerous convoluted lobes [145]. The lumen of lobes carry two types of cells i.e. hemocytoblasts and leukoblasts which varies in their cytoplasmic and nuclear content. The former one is larger having folded nucleus whereas later one carries larger nucleus [146]. There are certain fixed hemocytes present in tissues like gills and salivary glands whereas wandering hemocytes are present in dermal region. The functional role is similar to the granulocytes i.e., phagocytosis and encapsulation. The hemocytes from *Octopus briareus* seem to form rapid clots and polymerisation in the connective tissue and are less phagocytic in nature [147]. Upon skin damage in lesser octopus, the hemocytes infiltrate to the site and fusiform cells aggregate form dermal plug which eventually closes the wound within 120hr. The antigen induced steroid secretion from optic gland is having a marked defensive role in *O. vulgaris* [148]. The plasma from three cephalopods (*O. myu*, *Sepioteuthis lessoniana*, and *Sepia officinalis*) were able to agglutinate red blood cells and bacteria implying presence of conserved lectin like molecules in serum [149]. These lectins help in agglutination, cell adhesion and opsonization. There are few bacteria which don't get agglutinate, perhaps get recognized by opsonins. There are few protease compounds present in serum of cephalopods which selectively allow the symbionts to multiply whereas inhibits microbes by up regulating expression of lysozyme, chitinase and Cathepsins like enzymes. Protease inhibitors constitute a major component of cephalopod hemolymph which readily binds and inactivates pathogen derived proteases. Such proteases i.e., alpha-2-macroglobulin were found in *O. vulgaris*, cuttlefish *S. officinalis*, *Loligo vulgaris* and *L. pealii*, whose expression increases after challenge with bacteria [150,151]. A potential antibacterial protease inhibitor called serpin has been detected from *O. ocellatus* which can inactivate trypsin, chymotrypsin and *E. coli*. The transcript also upregulated upon challenge with *V. anguillarum* [152]. The expressed sequence tag library analysis of *Euprymna scolopes*, also known as the Hawaiian bobtail squid, has revealed several transcripts of Toll/NF-kappaB phosphorylation cascade were differentially expressed in light organ in response to *V. fischeri* [153]. However, pathway analysis PRR mediated immunity is obscured in cephalopods.

Crustacea

Among crustaceans' shrimp, crayfish, lobster and crabs' immune system is well documented because of their high demand for human consumption. Similar to molluscan, the crustaceans solely depend on innate defence mechanisms comprising cellular components i.e., hemocytes carrying pathogen recognition receptors, humoral components i.e., cytokines, lectins, AMPs, immunoglobulin, acute phase proteins. The hemocytes are involved in various functions like phagocytosis, apoptosis, encapsulation and nodule formation whereas humoral compounds act as immune effectors and antimicrobials.

There are three basic hemocytes based on their size and nuclear/cytoplasmic ratio i.e., hyaline cells, granular and semi granular cells. Hepatopancreas in shrimp and crab is known to be the hematopoietic organ and site of most immune gene expression. In shore crab *Carcinus maenas*, hyaline cells which are smallest cells with few cytoplasmic granules found highly phagocytic compare to other cell types whereas it is semi granular and granular cells mostly phagocytose particulate antigens in *Penaeus monodon* and *Macrobrachium rosenbergii*. In the same animal species, phagocytic cells may vary based on their response to different antigen varieties i.e., semi granular cells engulf White spot syndrome virus in red claw crayfish whereas both semi granular and granular response to *E. coli*. Apoptosis is another defensive response of hemocytes during infection, mostly viral infection. Upon challenge with WSSV, granulocytes of *Fenneropenaeus chinensis* undergo apoptosis in order to restrict spread of virus whereas hyaline cells are not involved in apoptosis [154]. The mature granulocytes of *P. monodon* proliferate into blast-like cells and migrate from hematopoietic organs towards the site

of viral infection, resembling an active involvement in antiviral activity [155]. Transcriptome analysis of challenges shrimp hemocytes has revealed, 3444 genes get differentially expressed and immune related genes are found involved in pattern recognition, related to signal transduction, antimicrobial peptides, ubiquitin mediated proteolysis, related to phagocytosis and proteases/protease inhibitor. Some of them are also involved in Neuroendocrine immunity [156]. Few activities of hemocytes resembles higher vertebrates i.e., formation of Extracellular traps with DNA, histone and other cell proteins [157]. Upon challenge with *E. coli*, the hemocytes of *Litopenaeus vannamei* throws its DNA and forms ECT with histone proteins in order to trap bacteria. Another major antimicrobial activity of hemocytes is PPO (Pro-Phenol Oxidase) system [158]. The PRR on hemocytes recognize the MAMPs i.e., LPS, β -Glucans, peptidoglycan and lectins. Upon triggering with microbial entry, it transfers the signals to serine proteases which converts zymogen PPO to phenol oxidase. A series of enzymes are involved and finally melanin is produced which is directly anti-microbial and indirectly forms nodules or encapsulates the antigen.

Humoral immunity includes AMPs, lectins, proteinase inhibitors, coagulation factors and various cytokines. Antimicrobial peptides or proteins are one of the major contributors of crustacean innate immunity which has a broad-spectrum antimicrobial activity i.e., the Penaeidin isolated from *L. vannamei* is both antibacterial and antifungal. Based on their amino acid sequence, these are categorized into 5 groups. Each of them has one conserved Cysteine rich domain and a variable proline rich domain. Based on their microbial target specificity the amino acid sequence varies. Penaeidin-4 is the most effective anti-fungal and anti-bacterial compound whereas penaeidin-3 is mostly active against gram positive bacteria [159]. Not all the humeral compounds are directly antimicrobial i.e., lectin, a major group of patterns recognizing glycoproteins found in crustaceans. They facilitate agglutination, endocytosis, phagocytosis, cell mediated immunity (encapsulation, nodulation, PPO and blood clotting). The C type lectin is the widespread group found in animals including crustaceans and their functional property reveals Ca^{+} dependent agglutination against gram positive and negative bacteria [160]. The shrimp lectins are also found to interact with viral envelope proteins suggesting a potential antiviral role in crustaceans [161]. Apart from lectins, different types of lysozymes have been discovered as potent bactericidal compounds. Lysozyme has the ability to cleave β -1,4 bonds in the peptidoglycan cell wall of gram-positive bacteria. Two types of lysozymes i.e., C-type and i-type have been detected from the EST database of *P. monodon*. These are expressed in all of the organs but mostly in hemocytes [162]. Silencing of Lysozyme gene by injecting dsRNA in *Marsupenaeus japonicus* revealed mass mortality without any external challenge but the hemolymph was carrying more bacterial load mostly *Vibrio* and *Pseudomonas* [163]. Hence it implies that lysozyme deprived penaeid shrimp could not eliminate bacterial infection and dies out of poor immunity. The cytokines, interleukins, interferons and platelet factors are also important antimicrobial and antiviral compounds present in crustaceans [164].

Several pattern recognition receptors have been discovered in crustaceans which are mostly classified based on choice of ligands, downstream signalling molecular cascade and effector molecules. It recognizes various PAMPs i.e., LPS, β -Glucan, peptidoglycan, dsRNA and viral proteins. Similar to previously discussed animals, crustaceans also carry TLR, RLR and NLR, where TLR refers to cell surface receptors and NLR and RLR are cytoplasmic in nature. There have been a total 11 TLR genes reported from *L. vannamei*, ubiquitously expressed in all tissues with highest expression in gill and hepatopancreas [165]. Such TLR genes are also reported from mud crab, horseshoe crab, crayfish and lobster, responding to various bacterial, viral and parasitic infection [166–169]. The receptors related to immunoglobulin super family i.e., down syndrome cell adhesion molecule (DSCAM) is believed to have potential similar to vertebrate Ig molecules. [170]. A single gene alternative splicing produces 18,000 different pattern specific receptors. It is

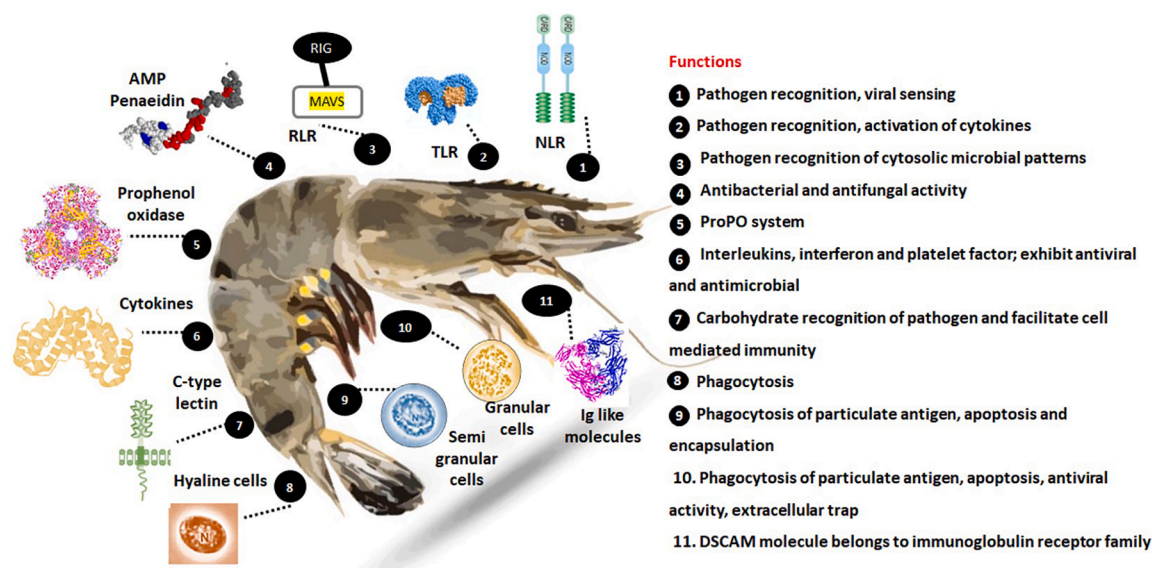


Fig. 5. Molecules involved in the innate immune responses of Crustaceans.

found both as transmembrane protein as well as secreted isoforms. The animals hemocyte which experimentally lacks DSCAM receptors show high susceptibility to bacterial infection, implying a strong role of DSCAM in pathogen recognition and receptor antigen signalling. The immune molecules associated with crustacean immunity as a model and their possible functions were depicted in Fig. 5.

Conclusion

In a timeline of billions of years ago, microbes were the only natural choice of animal unless the multicellular were evolved. Over the period when multicellular animals grew up, the microbes never left behind and there continues the race of two major dynasties. Some early animals i.e., sponge and corals contribute a major proportion of the body to symbiotic microbes which is a strategy to reduce the pressure of microbes' race. In price of living space and nourishment, the symbionts provide a passive immunity to animals. The group of symbiotic microbes are carried by most of the animals but the quantity has been reduced as the animal's own immunity has evolved to an extent which gives protection against a vast range of pathogens. Though invertebrates lack a B or T cell mediated specific immunity, the diverse isoforms of pattern recognition receptors i.e. TLR, NLR, RLR, DSCAM fulfill the requirement of antigen signalling as transmembrane as well as cytoplasmic receptors. Ancient animals including plants get exposure mediated genetic acquisition of effector molecular traits which supply a variable isoforms of pattern recognition receptors for immediate recognition and elimination of invading pathogen. This phenomenon of acquired immune memory of invertebrates is actually a consequence of host-microbe co-evolution where both pathogen and animal immunity get evolved in order to evade each other.

CRediT authorship contribution statement

R. Bharathi Rathinam: Writing – original draft. **Arpit Acharya:** Writing – review & editing. **A. Jancy Robina:** Writing – review & editing, Writing – original draft. **Husne Banu:** Writing – review & editing, Validation, Supervision, Conceptualization. **Gayatri Tripathi:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] J.T. Chun, F. Vasilev, N. Limatola, L. Santella, Fertilization in starfish and sea urchin: roles of actin, *Marine Org. Model Syst. Biol. Med.* (2018) 33–47.
- [2] C. Chasserat, F. Clarac, Basic processes of locomotor coordination in the rock lobster, *Biol. Cybern.* 55 (2) (1986) 171–185.
- [3] S. Hyodo, K. Kakumura, W. Takagi, K. Hasegawa, Y. Yamaguchi, Morphological and functional characteristics of the kidney of cartilaginous fishes: with special reference to urea reabsorption, *Am. J. Physiol.-Regul., Integr. Compar. Physiol.* 307 (12) (2014) R1381–R1395.
- [4] H.L. Pratt, J.C. Carrier, A review of elasmobranch reproductive behavior with a case study on the nurse shark, *Ginglymostoma cirratum*, *Environ. Biol. Fishes* 60 (1) (2001) 157–188.
- [5] N.J. Marra, M.J. Stanhope, N.K. Jue, M. Wang, Q. Sun, P.P. Bitar, V.P. Richards, A. Komissarov, M. Rayko, S. Kliver, B.J. Stanhope, White shark genome reveals ancient elasmobranch adaptations associated with wound healing and the maintenance of genome stability, *Proc. Natl. Acad. Sci.* 116 (10) (2019) 4446–4455.
- [6] C.R. Wilkinson, R. Garrone, J. Vacelet, Marine sponges discriminate between food bacteria and bacterial symbionts: electron microscope radioautography and in situ evidence, *Proc. R. Soc. London. Ser. B. Biol. Sci.* 220 (1221) (1984) 519–528.
- [7] L. Pita, M.P. Hoepfner, M. Ribes, U. Hentschel, Differential expression of immune receptors in two marine sponges upon exposure to microbial-associated molecular patterns, *Sci. Rep.* 8 (1) (2018) 1–15.
- [8] L. Pita, L. Rix, B.M. Slaby, A. Franke, U. Hentschel, The sponge holobiont in a changing ocean: from microbes to ecosystems, *Microbiome* 6 (1) (2018) 1–18.
- [9] S. Sun, C.B. Canning, K. Bhargava, X. Sun, W. Zhu, N. Zhou, Y. Zhang, K. Zhou, Polybrominated diphenyl ethers with potent and broad spectrum antimicrobial activity from the marine sponge *Dysidea*, *Bioorg. Med. Chem. Lett.* 25 (10) (2015) 2181–2183.
- [10] T. Wakimoto, Y. Egami, I. Abe, Calyculin: nature's way of making the sponge-derived cytotoxin, *Nat. Prod. Rep.* 33 (6) (2016) 751–760.
- [11] D.T. Youssef, L.A. Shaala, G.A. Mohamed, J.M. Badr, F.H. Bamanie, S.R. Ibrahim, Theonellamide G, a potent antifungal and cytotoxic bicyclic glycopeptide from the Red Sea marine sponge *Theonella swinhoei*, *Mar. Drugs* 12 (4) (2014) 1911–1923.
- [12] X. Bai, Y. Liu, H. Wang, H. Zhang, Natural products from the marine sponge subgenus *reniera*, *Molecules* 26 (4) (2021) 1097.
- [13] H. Horn, B.M. Slaby, M.T. Jahn, K. Bayer, L. Moitinho-Silva, F. Förster, U. R. Abdelmohsen, U. Hentschel, An enrichment of CRISPR and other defense-related features in marine sponge-associated microbial metagenomes, *Front. Microbiol.* 7 (2016) 1751.

- [14] C. Loureiro, A. Galani, A. Gavrilidou, M. Chaib de Mares, J. van der Oost, M. H. Medema, D. Sipkema, Comparative metagenomic analysis of biosynthetic diversity across sponge microbiomes highlights metabolic novelty, conservation, and diversification, *mSystems*. 7 (4) (2022) e00357-22.
- [15] N.S. Webster, R.E. Cobb, A.P. Negri, Temperature thresholds for bacterial symbiosis with a sponge, *ISME J.* 2 (8) (2008) 830–842.
- [16] P. Willenz, G. Van De Vyver, Endocytosis of latex beads by the exopinacoderm in the fresh water sponge *Ephydatia fluviatilis*: an in vitro and in situ study in SEM and TEM, *J. Ultrastruct. Res.* 79 (3) (1982) 294–306.
- [17] T. Humphreys, Rapid allogeneic recognition in the marine sponge *Microciona prolifera*: implications for evolution of immune recognition, *Ann. N. Y. Acad. Sci.* 712 (1) (1994) 342–345.
- [18] S. Pahler, B. Blumbach, I. Müller, W.E. Müller, Putative multiadhesive protein from the marine sponge *Geodia cydonium*: cloning of the cDNA encoding a fibronectin-, an SRCR-, and a complement control protein module, *J. Exp. Zool.* 282 (3) (1998) 332–343.
- [19] J. Gardères, M.L. Bourguet-Kondracki, B. Hamer, R. Batel, H.C. Schröder, W. E. Müller, Porifera lectins: diversity, physiological roles and biotechnological potential, *Mar. Drugs* 13 (8) (2015) 5059–5101.
- [20] D.N. Marques, A.S. de Almeida, A.R. de Oliveira Sousa, R. Pereira, A.L. Andrade, R.P. Chaves, R.F. Carneiro, M.A. de Vasconcelos, L.G. do Nascimento-Neto, U. Pinheiro, P.A. Videira, Antibacterial activity of a new lectin isolated from the marine sponge *Chondrilla caribensis*, *Int. J. Biol. Macromol.* 109 (2018) 1292–1301.
- [21] M.E. Gauthier, L. Du Pasquier, B.M. Degnan, The genome of the sponge *Amphimedon queenslandica* provides new perspectives into the origin of Toll-like and interleukin 1 receptor pathways, *Evol. Dev.* 12 (5) (2010) 519–533.
- [22] M. Wiens, M. Korzhnev, S. Perović-Ottstadt, B. Luthringer, D. Brandt, S. Klein, W. E. Müller, Toll-like receptors are part of the innate immune defense system of sponges (Demospongiae: porifera), *Mol. Biol. Evol.* 24 (3) (2007) 792–804.
- [23] B. Yuen, J.M. Bayes, S.M. Degnan, The characterization of sponge NLRs provides insight into the origin and evolution of this innate immune gene family in animals, *Mol. Biol. Evol.* 31 (1) (2014) 106–120.
- [24] S.M. Degnan, The surprisingly complex immune gene repertoire of a simple sponge, exemplified by the NLR genes: a capacity for specificity? *Dev. Compar. Immunol.* 48 (2) (2015) 269–274.
- [25] T. Ryu, L. Seridi, L. Moitinho-Silva, M. Oates, Y.J. Liew, C. Mavromatis, X. Wang, A. Haywood, F.F. Lafi, M. Kupresanin, R. Sougrat, Hologenome analysis of two marine sponges with different microbiomes, *BMC Genomics* 17 (1) (2016) 1–11.
- [26] L.L. Moroz, K.M. Kocot, M.R. Citarella, S. Dosung, T.P. Norekian, I. S. Povolotskaya, A.P. Grigorenko, C. Dailey, E. Berezikov, K.M. Buckley, A. Pitsyn, The ctenophore genome and the evolutionary origins of neural systems, *Nature* 510 (7503) (2014) 109–114.
- [27] S. Bolte, O. Roth, E.E. Philipp, J. Saphörster, P. Rosenstiel, T.B. Reusch, Specific immune priming in the invasive ctenophore *Mnemiopsis leidyi*, *Biol. Lett.* 9 (6) (2013) 20130864.
- [28] N. Traylor-Knowles, L.E. Vandepas, W.E. Browne, Still enigmatic: innate immunity in the ctenophore *Mnemiopsis leidyi*, *Integr. Comp. Biol.* 59 (4) (2019) 811–818.
- [29] C. Daniels, M. Breitbart, Bacterial communities associated with the ctenophores *Mnemiopsis leidyi* and *Beroë ovata*, *FEMS Microbiol. Ecol.* 82 (1) (2012) 90–101.
- [30] L.E. Vandepas, C. Stefani, N. Traylor-Knowles, F.W. Goetz, W.E. Browne, A. Lacy-Hulbert, Ctenophore immune cells produce chromatin traps in response to pathogens and NADPH-independent stimulus, *bioRxiv*. (2020).
- [31] S. Adhikary, S.P. Hui, The loss of regeneration competency in the animal kingdom at the expense of immunity: a journey in retrospect, *Brain Behav. Immun.* 94 (2021) 8–10.
- [32] E. Rosenberg, O. Koren, L. Reshef, R. Efrony, I. Zilber-Rosenberg, The role of microorganisms in coral health, disease and evolution, *Nat. Rev. Microbiol.* 5 (5) (2007) 355–362.
- [33] C.J. Krediet, K.B. Ritchie, M. Cohen, E.K. Lipp, K.P. Sutherland, M. Teplitski, Utilization of mucus from the coral *Acropora palmata* by the pathogen *Serratia marcescens* and by environmental and coral commensal bacteria, *Appl. Environ. Microbiol.* 75 (12) (2009) 3851–3858.
- [34] R. Rajasabapathy, S.C. Ghadi, B. Manikandan, C. Mohandass, A. Surendran, S. G. Dastager, R.A. James, Antimicrobial profiling of coral reef and sponge associated bacteria from southeast coast of India, *Microb. Pathog.* 141 (2020) 103972.
- [35] J. Nissimov, E. Rosenberg, C.B. Munn, Antimicrobial properties of resident coral mucus bacteria of *Oculina patagonica*, *FEMS Microb. Lett.* 292 (2) (2009) 210–215.
- [36] M. Shnit-Orland, A. Sivan, A. Kushmaro, Antibacterial activity of *Pseudoalteromonas* in the coral holobiont, *Microb. Ecol.* 64 (2012) 851–859.
- [37] K.B. Ritchie, G.W. Smith, Preferential carbon utilization by surface bacterial communities from water mass, normal, and white-band diseased *Acropora*, *Mol. Mar. Biol. Biotechnol.* 4 (1995) 345–352.
- [38] E.C.E. Kvennefors, E. Sampayo, C. Kerr, G. Vieira, G. Roff, A.C. Barnes, Regulation of bacterial communities through antimicrobial activity by the coral holobiont, *Microb. Ecol.* 63 (2012) 605–618.
- [39] N.N. Rosic, M. Pernice, S. Dove, S. Dunn, O. Hoegh-Guldberg, Gene expression profiles of cytosolic heat shock proteins Hsp70 and Hsp90 from symbiotic dinoflagellates in response to thermal stress: possible implications for coral bleaching, *Cell Stress Chaperones* 16 (1) (2011) 69–80.
- [40] M.K. DeSalvo, et al., Differential gene expression during thermal stress and bleaching in the Caribbean coral *Montastraea faveolata*, *Mol. Ecol.* 17 (17) (2008) 3952–3971.
- [41] E. Meyer, G.V. Aglyamova, M.V. Matz, Profiling gene expression responses of coral larvae (*Acropora millepora*) to elevated temperature and settlement inducers using a novel RNA-Seq procedure, *Mol. Ecol.* 20 (17) (2011) 3599–3616.
- [42] J. Vidal-Dupiol, O. Ladrière, D. Destoumieux-Garzon, P.E. Sautière, A. L. Meistertzheim, E. Tambutté, S. Tambutté, D. Duval, L. Fouré, M. Adjerdoud, G. Mitta, Innate immune responses of a scleractinian coral to vibriosis, *J. Biol. Chem.* 286 (25) (2011) 22688–22698.
- [43] B. Mason, I. Cooke, A. Moya, R. Augustin, M.F. Lin, N. Satoh, T.C.G. Bosch, D. G. Bourne, D.C. Hayward, N. Andrade, S. Forêt, AmAMP1 from *Acropora millepora* and damicornin define a family of coral-specific antimicrobial peptides related to the Shk toxins of sea anemones, *Dev. Compar. Immunol.* 114 (2021) 103866.
- [44] C.A. Ospina, A.D. Rodríguez, E. Ortega-Barria, T.L. Capson, Briarellins J–P and polyanthellin A: new eunicellin-based diterpenes from the gorgonian coral *Briareum polyanthes* and their antimalarial activity, *J. Nat. Prod.* 66 (3) (2003) 357–363.
- [45] C.A. Ospina, A.D. Rodríguez, J.A. Sánchez, E. Ortega-Barria, T.L. Capson, A. M. Mayer, Caucanolides A–F, Unusual Antiplasmodial Constituents from a Colombian Collection of the Gorgonian Coral *Pseudopterogorgia bipinnata*, *J. Nat. Prod.* 68 (10) (2005) 1519–1526.
- [46] C.A. Ospina, A.D. Rodríguez, H. Zhao, R.G. Raptis, Bipinnapterolide B, a bioactive oxapolycyclic diterpene from the Colombian gorgonian coral *Pseudopterogorgia bipinnata*, *Tetrahedron. Lett.* 48 (42) (2007) 7520–7523.
- [47] J.L. Shapo, P.D. Moeller, S.B. Galloway, Antimicrobial activity in the common seawhip, *Leptogorgia virgulata* (Cnidaria: gorgonaceae), *Compar. Biochem. Physiol. Part B: Biochem. Mol. Biol.* 148 (1) (2007) 65–73.
- [48] K. Kim, Antimicrobial activity in gorgonian corals (Coelenterata, Octocorallia), *Coral Reefs*. 13 (2) (1994) 75–80.
- [49] K. Kim, C.D. Harvell, P.D. Kim, G.W. Smith, S.M. Merkel, Fungal disease resistance of Caribbean sea fan corals (*Gorgonia* spp.), *Mar. Biol.* 136 (2) (2000) 259–267.
- [50] A. Meszaros, C.H. Bigger, Qualitative and quantitative study of wound healing processes in the coelenterate, *Plexaurella fusifera*: spatial, temporal, and environmental (light attenuation) influences, *J. Invert. Path.* 73 (1999) 321–331.
- [51] L.D. Mydlarz, S.F. Holthouse, E.C. Peters, C.D. Harvell, Cellular responses in sea fan corals: granular amoebocytes react to pathogen and climate stressors, *PLoS one* 3 (3) (2008) e1811.
- [52] C.T. Olano, C.H. Bigger, Phagocytic activities of the gorgonian coral *Swiftia exserta*, *J. Invertebr. Pathol.* 76 (3) (2000) 176–184.
- [53] K. Mullen, E.C. Peters, C.D. Harvell, Coral resistance to disease, in: E. Rosenberg, Y. Loya (Eds.), *Coral Health and Disease*, Springer-Verlag, New York, 2004, pp. 377–399.
- [54] B. Vargas-Angel, E.C. Peters, E. Kramarsky-Winter, D.S. Gilliam, R.E. Dodge, Cellular reactions to sedimentation and temperature stress in the Caribbean coral *Montastraea cavernosa*, *J. Invert. Path.* 95 (2007) 140–145.
- [55] C. Shinzato, E. Shoguchi, T. Kawashima, M. Hamada, K. Hisata, M. Tanaka, M. Fujie, M. Fujiwara, R. Koyanagi, T. Ikuta, A. Fujiyama, Using the *Acropora digitifera* genome to understand coral responses to environmental change, *Nature* 476 (7360) (2011) 320–323.
- [56] D.J. Miller, G. Hemmrich, E.E. Ball, D.C. Hayward, K. Khalturin, N. Funayama, T. C. Bosch, The innate immune repertoire in Cnidaria-ancestral complexity and stochastic gene loss, *Genome Biol.* 8 (2007) 1–13.
- [57] M. Hamada, E. Shoguchi, C. Shinzato, T. Kawashima, D.J. Miller, N. Satoh, The complex NOD-like receptor repertoire of the coral *Acropora digitifera* includes novel domain combinations, *Mol. Biol. Evol.* 30 (1) (2013) 167–176.
- [58] B.A. Dimos, C.C. Butler, C.A. Ricci, N.J. MacKnight, L.D. Mydlarz, Responding to threats both foreign and domestic: NOD-like receptors in corals, *Integr. Comp. Biol.* 59 (4) (2019) 819–829.
- [59] A.Z. Poole, V.M. Weis, TIR-domain-containing protein repertoire of nine anthozoan species reveals coral-specific expansions and uncharacterized proteins, *Dev. Compar. Immunol.* 46 (2) (2014) 480–488.
- [60] E.C.E. Kvennefors, W. Leggat, O. Hoegh-Guldberg, B.M. Degnan, A.C. Barnes, An ancient and variable mannose-binding lectin from the coral *Acropora millepora* binds both pathogens and symbionts, *Dev. Compar. Immunol.* 32 (12) (2008) 1582–1592.
- [61] Y. Endo, M. Takahashi, T. Fujita, Lectin complement system and pattern recognition, *Immunobiology* 211 (4) (2006) 283–293.
- [62] Y. Zhu, S. Thangamani, B. Ho, J.L. Ding, The ancient origin of the complement system, *EMBO J.* 24 (2) (2005) 382–394.
- [63] H.-M. Shen, S. Pervaiz, TNF receptor superfamily-induced cell death: redox-dependent execution, *FASEB J.* 20 (10) (2006) 1589–1598.
- [64] M.L. Hayes, R.I. Eytan, M.E. Hellberg, High amino acid diversity and positive selection at a putative coral immunity gene (tachylectin-2), *BMC Evol. Biol.* 10 (1) (2010) 1–15.
- [65] D.L. Brower, S.M. Brower, D.C. Hayward, E.E. Ball, Molecular evolution of integrins: genes encoding integrin β subunits from a coral and a sponge, *Proc. Natl. Acad. Sci.* 94 (17) (1997) 9182–9187.
- [66] B.A. Knack, A. Iguchi, C. Shinzato, D.C. Hayward, E.E. Ball, D.J. Miller, Unexpected diversity of cnidarian integrins: expression during coral gastrulation, *BMC Evol. Biol.* 8 (1) (2008) 1–11.
- [67] E. Rosenberg, L. Falkovitz, The Vibrio shiloi/Oculina patagonica model system of coral bleaching, *Annu. Rev. Microbiol.* 58 (2004) 143–159.
- [68] L. Reshef, O. Koren, Y. Loya, I. Zilber-Rosenberg, E. Rosenberg, The coral probiotic hypothesis, *Environ. Microbiol.* 8 (12) (2006) 2068–2073.
- [69] R.W. Buddemeier, D.G. Fautin, Coral bleaching as an adaptive mechanism, *Bioscience* 43 (5) (1993) 320–326.

- [70] H. Nagai, K. Takuwa, M. Nakao, E. Ito, M. Miyake, M. Noda, T. Nakajima, Novel proteinaceous toxins from the box jellyfish (sea wasp) *Carybdea rastoni*, *Biochem. Biophys. Res. Commun.* 275 (2) (2000) 582–588.
- [71] H. Nagai, K. Takuwa-Kuroda, M. Nakao, N. Oshiro, S. Iwanaga, T. Nakajima, A novel protein toxin from the deadly box jellyfish (sea wasp, Habu-kurage) *Chiropsalmus quadrigatus*, *Biosci. Biotechnol. Biochem.* 66 (1) (2002) 97–102.
- [72] D.L. Brinkman, X. Jia, J. Potriquet, D. Kumar, D. Dash, D. Kvaskoff, J. Mulvenna, Transcriptome and venom proteome of the box jellyfish *Chironex fleckeri*, *BMC Genomics* 16 (1) (2015) 1–15.
- [73] R. Li, H. Yu, Y. Yue, S. Liu, R. Xing, X. Chen, P. Li, Combined proteomics and transcriptomics identifies sting-related toxins of jellyfish *Cyanea nozakii*, *J. Proteomics* 148 (2016) 57–64.
- [74] W. Liu, F. Mo, G. Jiang, H. Liang, C. Ma, T. Li, L. Zhang, L. Xiong, G.L. Mariottini, J. Zhang, L. Xiao, Stress-induced mucus secretion and its composition by a combination of proteomics and metabolomics of the jellyfish *Aurelia coerulesa*, *Mar. Drugs* 16 (9) (2018) 341.
- [75] K.M. Kramar, T. Tinta, D. Lucić, A. Malej, V. Turk, Bacteria associated with moon jellyfish during bloom and post-bloom periods in the Gulf of Trieste (northern Adriatic), *PLoS One* 14 (1) (2019) e0198056.
- [76] N. Weiland-Brauer, S.C. Neulinger, N. Pinnow, S. Künzel, J.F. Baines, R. A. Schmitz, Composition of bacterial communities associated with *Aurelia aurita* changes with compartment, life stage, and population, *Appl. Environ. Microbiol.* 81 (17) (2015) 6038–6052.
- [77] T.V. Ovchinnikova, S.V. Balandin, G.M. Aleshina, A.A. Tagaev, Y.F. Leonova, E. D. Krasnodomskiy, A.V. Men'shenin, V.N. Kokryakov, Aurelin, a novel antimicrobial peptide from jellyfish *Aurelia aurita* with structural features of defensins and channel-blocking toxins, *Biochem. Biophys. Res. Commun.* 348 (2) (2006) 514–523.
- [78] H.S. Moon, Y.K. Kim, M.H. Lee, N.Y. Yoon, D.S. Lee, H.D. Yoon, J.K. Seo, N. G. Park, Isolation and purification of an antimicrobial material from the jellyfish *Nemopilema nomurai*, Korean J. Fisher. Aquat. Sci. 44 (5) (2011) 478–483.
- [79] L. Stabili, L. Rizzo, F.P. Fanizzi, F. Angilè, L. Del Coco, C.R. Girelli, S. Lomartire, S. Piraino, L. Basso, The jellyfish *Rhizostoma pulmo* (Cnidaria): biochemical composition of ovaries and antibacterial lysozyme-like activity of the oocyte lysate, *Mar. Drugs* 17 (1) (2019) 17.
- [80] M.J. Abrams, L. Goentoro, Symmetrization in jellyfish: reorganization to regain function, and not lost parts, *Zoology* 119 (1) (2016) 1–3.
- [81] U. Technau, S. Rudd, P. Maxwell, P.M. Gordon, M. Saina, L.C. Grasso, D.J. Miller, Maintenance of ancestral complexity and non-metazoan genes in two basal cnidarians, *TRENDS in Genetics* 21 (12) (2005) 633–639.
- [82] A. Kusserow, K. Pang, C. Sturm, M. Hrouda, J. Lentfer, H.A. Schmidt, T. W. Holstein, Unexpected complexity of the Wnt gene family in a sea anemone, *Nature* 433 (7022) (2005) 156–160.
- [83] R. Lubbock, Clone-specific cellular recognition in a sea anemone, *Proceedings of the National Academy of Sciences* 77 (11) (1980) 6667–6669.
- [84] K.P. Sebans, Agonistic behavior in the intertidal sea anemone *Anthopleura xanthogramma*, *The Biological Bulletin* 166 (3) (1984) 457–472.
- [85] J.J. Brennan, J.L. Messerschmidt, L.M. Williams, B.J. Matthews, M. Reynoso, T. D. Gilmore, Sea anemone model has a single Toll-like receptor that can function in pathogen detection, NF- κ B signal transduction, and development, *Proc. Natl. Acad. Sci.* 114 (47) (2017) E10122–E10131.
- [86] K.M. Mansfield, N.M. Carter, L. Nguyen, P.A. Cleves, A. Alshanbayeva, L. M. Williams, T.D. Gilmore, Transcription factor NF- κ B is modulated by symbiotic status in a sea anemone model of cnidarian bleaching, *Sci. Rep.* 7 (1) (2017) 16025.
- [87] S.R. Margolis, P.A. Dietzen, B.M. Hayes, S.C. Wilson, B.C. Remick, S. Chou, R. E. Vance, The cyclic dinucleotide 2' 3'-cGAMP induces a broad antibacterial and antiviral response in the sea anemone *Nematostella vectensis*, *Proc. Natl. Acad. Sci.* 118 (51) (2021) e2109022118.
- [88] A.Z. Poole, S.A. Kitchen, V.M. Weis, The role of complement in cnidarian-dinoflagellate symbiosis and immune challenge in the sea anemone *Aiptasia pallida*, *Front. Microbiol.* 7 (2016) 519.
- [89] A. Kimura, E. Sakaguchi, M. Nonaka, Multi-component complement system of Cnidaria: C3, Bf, and MASP genes expressed in the endodermal tissues of a sea anemone, *Nematostella vectensis*, *Immunobiology* 214 (3) (2009) 165–178.
- [90] E.F. Neubauer, A.Z. Poole, O. Detournay, V.M. Weis, S.K. Davy, The scavenger receptor repertoire in six cnidarian species and its putative role in cnidarian-dinoflagellate symbiosis, *PeerJ* 4 (2016) e2692.
- [91] E.F. Neubauer, A.Z. Poole, P. Neubauer, O. Detournay, K. Tan, S.K. Davy, V. M. Weis, A diverse host thrombospondin-type-1 repeat protein repertoire promotes symbiont colonization during establishment of cnidarian-dinoflagellate symbiosis, *Elife* 6 (2017) e24494.
- [92] S. Perez, V. Weis, Nitric oxide and cnidarian bleaching: an eviction notice mediates breakdown of a symbiosis, *J. Exp. Biol.* 209 (14) (2006) 2804–2810.
- [93] M.C. Chen, Y.M. Cheng, P.J. Sung, C.E. Kuo, L.S. Fang, Molecular identification of Rab7 (ApRab7) in *Aiptasia pulchella* and its exclusion from phagosomes harboring zooxanthellae, *Biochem. Biophys. Res. Commun.* 308 (3) (2003) 586–595.
- [94] S.R. Dunn, W.S. Phillips, J.W. Spatafora, D.R. Green, V.M. Weis, Highly conserved caspase and Bcl-2 homologues from the sea anemone *Aiptasia pallida*: lower metazoans as models for the study of apoptosis evolution, *J. Mol. Evol.* 63 (2006) 95–107.
- [95] J.P. Rast, L.C. Smith, M. Loza-Coll, T. Hibino, G.W. Litman, Genomic insights into the immune system of the sea urchin, *Science* (1979) 314 (5801) (2006) 952–956.
- [96] H. Ito, T. Tomooka, N. Sakai, H. Yu, Y. Higashino, K. Fujii, T. Masuyama, A. Kitabatake, T. Minamino, Lack of myocardial perfusion immediately after successful thrombolysis. A predictor of poor recovery of left ventricular function in anterior myocardial infarction, *Circulation* 85 (5) (1992) 1699–1705.
- [97] M.A. Yui, C.J. Bayne, Echinoderm immunology: bacterial clearance by the sea urchin *Strongylocentrotus purpuratus*, *Biol. Bull.* 165 (2) (1983) 473–486, <https://doi.org/10.2307/1541213>.
- [98] H.Y. Chou, C.M. Lun, L.C. Smith, SpTransformer proteins from the purple sea urchin opsonize bacteria, augment phagocytosis, and retard bacterial growth, *PLoS One* 13 (5) (2018) e0196890, <https://doi.org/10.1371/journal.pone.0196890>.
- [99] L.C. Smith, J.P. Rast, V. Brockton, D.P. Terwilliger, S.V. Nair, K.M. Buckley, A. J. Majeske, The sea urchin immune system, *Invertebrate Surv. J.* 3 (1) (2006) 25–39.
- [100] L.C. Smith, T.S. Hawley, J.H. Henson, A.J. Majeske, M. Oren, B. Rosental, Methods for collection, handling, and analysis of sea urchin coelomocytes, *Methods Cell Biol.* 150 (2019) 357–389, <https://doi.org/10.1016/bbs.mcb.2018.11.009>.
- [101] M. Taguchi, S. Tsutsui, O. Nakamura, Differential count and time-course analysis of the cellular composition of coelomocyte aggregate of the Japanese sea cucumber *Apostichopus japonicus*, *Fish Shellfish Immunol.* 58 (2016) 203–209, <https://doi.org/10.1016/j.fsi.2016.06.060>.
- [102] L.C. Smith, J. Ghosh, K.M. Buckley, L.A. Clow, N.M. Dheilly, T. Haug, J. H. Henson, C. Li, C.M. Lun, A.J. Majeske, V. Matraga, Echinoderm immunity, *Invertebrate Immun.* (2010) 260–301, https://doi.org/10.1007/978-1-4419-8059-5_14.
- [103] F.S. Chia, J. Xing, Echinoderm coelomocytes, *Zool. Stud.-Taipei* 35 (1996) 231–254.
- [104] A. Pinsino, M.C. Thorndyke, V. Matraga, Coelomocytes and post-traumatic response in the common sea star *Asterias rubens*, *Cell Stress. Chaperones* 12 (4) (2007) 331, <https://doi.org/10.1379/CSC-288.1>.
- [105] B.J. Hillier, V.D. Vacquier, Amassin, an olfactomedin protein, mediates the massive intercellular adhesion of sea urchin coelomocytes, *J. Cell Biol.* 160 (4) (2003) 597–604, <https://doi.org/10.1083/jcb.200210053>.
- [106] P.S. Asha, P. Muthiah, Spawning and larval rearing of sea cucumber *Holothuria* (Theelothuria) spinifera Theel, *SPC Beche-de-mer, Inf. Bull.* 16 (2) (2002) 11–15.
- [107] C. Canicatti, G. D'Ancona, Cellular aspects of *Holothuria polii* immune response, *J. Invertebr. Pathol.* 53 (2) (1989) 152–158.
- [108] B. Lepenies, R. Lang, Editorial: lectins and their ligands in shaping immune responses, *Front. Immunol.* 10 (2019) 2379.
- [109] T. Hatakeyama, H. Kohzaki, H. Nagatomo, N. Yamasaki, Purification and characterization of four Ca²⁺-dependent lectins from the marine invertebrate, *Cucumaria echinata*, *J. Biochem.* 116 (1) (1994) 209–214.
- [110] D. Schillaci, M.G. Cusimano, V. Consolo, R. Saletti, D. Russo, M. Vazzana, M. Vitale, V. Arizza, Immune mediators of sea-cucumber *Holothuria tubulosa* (Echinodermata) as source of novel antimicrobial and anti-staphylococcal biofilm agents, *AMB Express* 3 (1) (2013) 1–10, <https://doi.org/10.1186/2191-0855-3-35>.
- [111] F. Wang, H. Yang, F. Gao, G. Liu, Effects of acute temperature or salinity stress on the immune response in sea cucumber, *Apostichopus japonicus*, *Compar. Biochem. Physiol. Part A: Mol. Integr. Physiol.* 151 (4) (2008) 491–498, <https://doi.org/10.1016/j.cbpa.2008.06.024>.
- [112] Q. Wang, C. Wang, C. Mu, H. Wu, L. Zhang, J. Zhao, A novel C-type lysozyme from *Mytilus galloprovincialis*: insight into innate immunity and molecular evolution of invertebrate C-type lysozymes, *PLoS One* 8 (6) (2013) e67469.
- [113] Z. Zhou, D. Sun, A. Yang, Y. Dong, Z. Chen, X. Wang, B. Wang, Molecular characterization and expression analysis of a complement component 3 in the sea cucumber (*Apostichopus japonicus*), *Fish Shellfish Immunol.* 31 (4) (2011) 540–547.
- [114] J. Xing, F.S. Chia, Opsonin-like molecule found in coelomic fluid of a sea cucumber, *Holothuria leucospilota*, *Mar. Biol.* 136 (2000) 979–986.
- [115] L.A. Clow, P.S. Gross, C.S. Shih, L.C. Smith, Expression of SpC3, the sea urchin complement component, in response to lipopolysaccharide, *Immunogenet.* 51 (12) (2000) 1021–1033, <https://doi.org/10.1007/s002510000233>.
- [116] L. Cong, X. Yang, X. Wang, M. Tada, M. Lu, H. Liu, B. Zhu, Characterization of an i-type lysozyme gene from the sea cucumber *Stichopus japonicus*, and enzymatic and nonenzymatic antimicrobial activities of its recombinant protein, *J. Biosci. Bioeng.* 107 (6) (2009) 583–588, <https://doi.org/10.1016/j.jbiosc.2009.01.016>.
- [117] F. Ramírez-Gómez, J.E. García-Arriarás, Echinoderm immunity, *Invertebr. Surv. J.* 7 (2) (2010) 211–220.
- [118] D. Jans, P. Dubois, M. Jangoux, Defensive mechanisms of holothuroids (Echinodermata): formation, role, and fate of intracoelomic brown bodies in the sea cucumber *Holothuria tubulosa*, *Cell Tissue Res.* 283 (1) (1995) 99–106, <https://doi.org/10.1007/s004410050517>.
- [119] Z. Xue, H. Li, X. Wang, X. Li, Y. Liu, J. Sun, C. Liu, A review of the immune molecules in the sea cucumber, *Fish Shellfish Immunol.* 44 (1) (2015) 1–11, <https://doi.org/10.1016/j.fsi.2015.01.026>.
- [120] M. Chiamonte, R. Russo, The echinoderm innate humoral immune response, *Italian J. Zool.* 82 (3) (2015) 300–308, <https://doi.org/10.1080/11250003.2015.1061615>.
- [121] D.A. Mogilenko, I.V. Kudryavtsev, S.V. Orlov, A.D. Kharasova, A.V. Polevshikov, Expression of the starfish complement component C3 gene homolog under the influence of bacterial lipopolysaccharide, *Mol. Biol. (N.Y.)* 44 (1) (2010) 67–76, <https://doi.org/10.1134/S0026893310010097>.
- [122] G. Coteur, D. Gillan, G. Joly, Field contamination of the starfish *Asterias rubens* by metals. Part 2: effects on cellular immunity, *Environ. Toxicol. Chem.* 22 (9) (2003) 2145–2151, 2003.

- [123] B. Danis, P. Wantier, R. Flammang, et al., Bioaccumulation and effects of PCBs and heavy metals in sea stars (*Asterias rubens*, L.) from the North Sea: a small scale perspective, *Sci. Total. Environ.* (2006).
- [124] G. Coteur, P. Gosselin, P. Wantier, Y. Chambost-Manciet, B. Danis, P. Pernet, M. Warnau, P. Dubois, Echinoderms as bioindicators, bioassays, and impact assessment tools of sediment-associated metals and PCBs in the North Sea, *Arch. Environ. Contam. Toxicol.* 45 (2) (2003) 190–202, <https://doi.org/10.1007/s00244-003-0199-x>.
- [125] H. Kamiya, K. Muramoto, R. Goto, M. Sakai, Lectins in the hemolymph of a starfish, *Asterina pectinifera*: purification and characterization, *Dev. Compar. Immunol.* 16 (2–3) (1992) 243–250, [https://doi.org/10.1016/0145-305X\(92\)90023-6](https://doi.org/10.1016/0145-305X(92)90023-6).
- [126] M.D. Candia-Carnevali, M.C. Thorndyke, V. Matranga, Regenerating echinoderms: a promise to understand stem cells potential. *Stem Cells in Marine Organisms*, Springer, Dordrecht, 2009, pp. 165–186, https://doi.org/10.1007/978-90-481-2767-2_7.
- [127] K. Kanungo, In vitro studies on the effects of cell-free coelomic fluid, calcium, and/or magnesium on clumping of coelomocytes of the sea star *Asterias forbesi* (Echinodermata: asteroidea), *Biol. Bull.* 163 (3) (1982) 438–452, <https://doi.org/10.2307/1541455>.
- [128] S.R. Margolis, P.A. Dietzen, B.M. Hayes, S.C. Wilson, B.C. Remick, S. Chou, R. E. Vance, The cyclic dinucleotide 2' 3'-cGAMP induces a broad antibacterial and antiviral response in the sea anemone *Nematostella vectensis*, *Proc. Natl. Acad. Sci.* 118 (51) (2021) e2109022118.
- [129] P. Heger, W. Zheng, A. Rottmann, K.A. Panfilio, T. Wiehe, The genetic factors of bilaterian evolution, *Elife* 9 (2020) e45530.
- [130] E.S. Loker, Macroevolutionary immunology: a role for immunity in the diversification of animal life, *Front. Immunol.* 3 (2012) 25.
- [131] Z.D.Z.I.S.L.A.W. Gliński, J. Jarosz, Molluscan immune defenses, *Arch. Immunol. Ther. Exp. (Warsz.)* 45 (2–3) (1997) 149–155.
- [132] J.H. Schultz, C.M. Adema, Comparative immunogenomics of molluscs, *Dev. Compar. Immunol.* 75 (2017) 3–15.
- [133] R.H. Baxter, A. Contet, K. Krueger, Arthropod innate immune systems and vector-borne diseases, *Biochem.* 56 (7) (2017) 907–918.
- [134] A. Kulkarni, S. Krishnan, D. Anand, S. Kokkattunivarthil Uthaman, S.K. Otta, I. Karunasagar, R. Kooloth Valappil, Immune responses and immunoprotection in crustaceans with special reference to shrimp, *Rev. Aquac.* 13 (1) (2021) 431–459.
- [135] C.B. Kim, S.Y. Moon, S.R. Gelder, W. Kim, Phylogenetic relationships of annelids, molluscs, and arthropods evidenced from molecules and morphology, *J. Mol. Evol.* 43 (3) (1996) 207–215.
- [136] N. Nantararat, Y. Tragoolpua, P. Gunama, Antibacterial activity of the mucus extract from the Giant African Snail (*Lissachatina fulica*) and golden apple snail (*Pomacea canaliculata*) against pathogenic bacteria causing skin diseases, *Trop. Nat. History* 19 (2) (2019) 103–112.
- [137] G.G. Martin, C.T. Oakes, H.R. Tousignant, H. Crabtree, R. Yamakawa, Structure and function of haemocytes in two marine gastropods, *Megathura crenulata* and *Aplysia californica*, *J. Molluscan Stud.* 73 (4) (2007) 355–365.
- [138] L. Donaghy, H.K. Hong, C. Lambert, H.S. Park, W.J. Shim, K.S. Choi, First characterisation of the populations and immune-related activities of hemocytes from two edible gastropod species, the disk abalone, *Haliotis discus discus* and the spiny top shell, *Turbo cornutus*, *Fish Shellfish Immunol.* 28 (1) (2010) 87–97.
- [139] B. Allam, D. Raftos, Immune responses to infectious diseases in bivalves, *J. Invertebr. Pathol.* 131 (2015) 121–136.
- [140] Y.T. Lau, L. Sussman, E.P. Espinosa, S. Katalay, B. Allam, Characterization of hemocytes from different body fluids of the eastern oyster *Crassostrea virginica*, *Fish Shellfish Immunol.* 71 (2017) 372–379.
- [141] S. Aladaileh, S.V. Nair, D. Birch, D.A. Raftos, Sydney rock oyster (*Saccostrea glomerata*) hemocytes: morphology and function, *J. Invertebr. Pathol.* 96 (1) (2007) 48–63.
- [142] L. Zhang, L. Li, Y. Zhu, G. Zhang, X. Guo, Transcriptome analysis reveals a rich gene set related to innate immunity in the Eastern oyster (*Crassostrea virginica*), *Marine Biotechnol.* 16 (1) (2014) 17–33.
- [143] A.M. Gorbushin, N.V. Iakovleva, Dscam receptors from mollusca: multiple isoforms are lacking, *Fish Shellfish Immunol.* 6 (34) (2013) 1653–1654.
- [144] H. Zhang, H. Wang, L. Wang, X. Song, J. Zhao, L. Qiu, L. Li, M. Cong, L. Song, A novel C-type lectin (Cflec-3) from *Chlamys farreri* with three carbohydrate-recognition domains, *Fish Shellfish Immunol.* 26 (5) (2009) 707–715.
- [145] M.G. Castillo, K.A. Salazar, N.R. Joffe, The immune response of cephalopods from head to foot, *Fish Shellfish Immunol.* 46 (1) (2015) 145–160.
- [146] L.A. Ford, Host defense mechanisms of cephalopods, *Annu. Rev. Fish Dis.* 2 (1992) 25–41.
- [147] R.R. Cowden, S.K. Curtis, Observations on living cells dissociated from the leukopoietic organ of *Octopus briareus*, *Exp. Mol. Pathol.* 19 (2) (1973) 178–185.
- [148] D. Froesch, Antigen-induced secretion in the optic gland of *Octopus vulgaris*, *Proc. R. Soc. Lond. B. Biol. Sci.* 205 (1160) (1979) 379–384.
- [149] W.S. Fisher, A.R. DiNuzzo, Agglutination of bacteria and erythrocytes by serum from six species of marine molluscs, *J. Invertebr. Pathol.* 57 (3) (1991) 380–394.
- [150] I.B. Thøgersen, G. Salvesen, F.H. Brucato, S.V. Pizzo, J.J. Enghild, Purification and characterization of an α -macroglobulin proteinase inhibitor from the mollusc *Octopus vulgaris*, *Biochem. J.* 285 (2) (1992) 521–527.
- [151] K. Vanhoorelbeke, A. Goossens, C. Gieles, G. Préaux, An α 2-macroglobulin-like proteinase inhibitor in the haemolymph of the Decapod cephalopod *Sepia officinalis*, *Arch. Physiol. Biochem.* (1994) 102.
- [152] X. Wei, J. Xu, J. Yang, X. Liu, R. Zhang, W. Wang, J. Yang, Involvement of a Serpin serine protease inhibitor (OoSerpin) from mollusc *Octopus ocellatus* in antibacterial response, *Fish Shellfish Immunol.* 42 (1) (2015) 79–87.
- [153] M.S. Goodson, M. Kojadinovic, J.V. Troll, T.E. Scheetz, T.L. Casavant, M. B. Soares, M.J. McFall-Ngai, Identifying components of the NF- κ B pathway in the beneficial *Euprymna scolopes-Vibrio fischeri* light organ symbiosis, *Appl. Environ. Microbiol.* 71 (11) (2005) 6934–6946.
- [154] C. Cui, Q. Liang, X. Tang, J. Xing, X. Sheng, W. Zhan, Differential apoptotic responses of hemocyte subpopulations to white spot syndrome virus infection in *Penaeus chinensis*, *Front. Immunol.* 11 (2020) 3174.
- [155] C.B.T. Van de Braak, M.H.A. Botterblom, E.A. Huisman, J.H.W.M. Rombout, W.P. W. Van der Knaap, Preliminary study on haemocyte response to white spot syndrome virus infection in black tiger shrimp *Penaeus monodon*, *Dis. Aquat. Org.* 51 (2) (2002) 149–155.
- [156] F. Wang, S. Li, J. Xiang, F. Li, Transcriptome analysis reveals the activation of neuroendocrine-immune system in shrimp hemocytes at the early stage of WSSV infection, *BMC Genomics* 20 (1) (2019) 1–14.
- [157] T.H. Ng, S.H. Chang, M.H. Wu, H.C. Wang, Shrimp hemocytes release extracellular traps that kill bacteria, *Dev. Compar. Immunol.* 41 (4) (2013) 644–651.
- [158] P. Amparyup, W. Charoensapsri, A. Tassanakajon, Phospholipase system and its role in shrimp immune responses against major pathogens, *Fish Shellfish Immunol.* 34 (4) (2013) 990–1001.
- [159] B.J. Cuthbertson, Y. Yang, E. Bachère, E.E. Büllsbach, P.S. Gross, A. Aumelas, Solution structure of synthetic penaeidin-4 with structural and functional comparisons with penaeidin-3, *J. Biol. Chem.* 280 (16) (2005) 16009–16018.
- [160] L. Cerenius, P. Jiravanichpaisal, H.P. Liu, I. Soderhall, Crustacean immunity, *Invertebr. Immun.* (2010) 239–259.
- [161] Z.Y. Zhao, Z.X. Yin, X.P. Xu, S.P. Weng, X.Y. Rao, Z.X. Dai, Y.W. Luo, G. Yang, Z. S. Li, H.J. Guan, S.D. Li, A novel C-type lectin from the shrimp *Litopenaeus vannamei* possesses anti-white spot syndrome virus activity, *J. Virol.* 83 (1) (2009) 347–356.
- [162] P. Supungul, V. Rimphanitchayakit, T. Aoki, I. Hirono, A. Tassanakajon, Molecular characterization and expression analysis of a c-type and two novel muramidase-deficient i-type lysozymes from *Penaeus monodon*, *Fish Shellfish Immunol.* 28 (3) (2010) 490–498.
- [163] A. Kaizu, F.F. Fagutao, H. Kondo, T. Aoki, I. Hirono, Functional analysis of C-type lysozyme in penaeid shrimp, *J. Biol. Chem.* 286 (52) (2011) 44344–44349.
- [164] Y. Chen, J. Cao, X. Zhang, The role of cytokine PF4 in the antiviral immune response of shrimp, *PLoS One* 11 (9) (2016) e0162954.
- [165] Y.J. Habib, H. Wan, Y. Sun, J. Shi, C. Yao, J. Lin, H. Ge, Y. Wang, Z. Zhang, Genome-wide identification of toll-like receptors in Pacific white shrimp (*Litopenaeus vannamei*) and expression analysis in response to *Vibrio parahaemolyticus* invasion, *Aquaculture* 532 (2021) 735996.
- [166] K.F. Clark, A.R. Acorn, S.J. Greenwood, A transcriptomic analysis of American lobster (*Homarus americanus*) immune response during infection with the bumper car parasite *Anophryoides haemophila*, *Dev. Compar. Immunol.* 40 (2) (2013) 112–122.
- [167] K.I. Inamori, S. Arikawa, S.I. Kawabata, A Toll-like receptor in horseshoe crabs, *Immunol. Rev.* 198 (1) (2004) 106–115.
- [168] J.F. Lan, S. Wei, Y.Q. Wang, Y.J. Dai, J.G. Tu, L.J. Zhao, X.C. Li, Q.W. Qin, N. Chen, L. Lin, PcToll3 was involved in anti-Vibrio response by regulating the expression of antimicrobial peptides in red swamp crayfish, *Procambarus clarkii*, *Fish Shellfish Immunol.* 57 (2016) 17–24.
- [169] R. Vidya, A. Paria, A. Deepika, K. Sreedharan, M. Makesh, C.S. Purushothaman, A. Chaudhari, P.G. Babu, K.V. Rajendran, Toll-like receptor of mud crab, *Scylla serrata*: molecular characterisation, ontogeny and functional expression analysis following ligand exposure, and bacterial and viral infections, *Mol. Biol. Rep.* 41 (10) (2014) 6865–6877.
- [170] D. Cooper, I. Eleftherianos, Memory and specificity in the insect immune system: current perspectives and future challenges, *Front. Immunol.* 8 (2017) 539.